Master's thesis

NTNU
Norwegian University of Science and Technology
Faculty of Information Technology and Electrical Engineering
Department of Electric Energy

Andreas Fosse Hansen and Jonas Henrik Pinderud

Operation of an EV parking lot subject to capacity-based grid tariffs offering grid services through demand response

Master's thesis in Energy and Environmental Engineering
Supervisor: Kasper Emil Thorvaldsen
June 2024

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Abstract

While the rapid increase in the adoption of Electric Vehicles (EVs) brings forth many environmental and economic benefits, it also presents new challenges for the power systems. To make the EVs a part of the solution and not a part of the problem, a flexible EV parking lot is investigated. The operation of the EV parking lot is optimized to reduce the cost of operation and peak consumption for the connected office building, while also offering grid-services through an incentive-based demand response program (IBDRP). Data from an existing office building and EV parking lot is used together with day-ahead spot price data.

To handle the short-term planning in the operation of the office building and connected EV parking lot, a Receding Horizon (RH) framework is implemented. The RH framework can handle the IBDRP activation signals which are given with varying notice time before the the parking lot shuts off the charging. The decomposition technique Stochastic Dual Dynamic Programming (SDDP) is used to provide future cost curves to the RH planning. The future cost curves are implemented to strategically account for the long-term implications of raising the monthly peak demand due to the demand charge cost. The combined model is tested on varying notice times, varying degrees of EV charging reduction in the IBDRP activation, and an implementation of Vehicle-to-grid (V2G) technology which can provide power back to the office building and/or power grid.

The model performance is evaluated under various notice times (0, 2, and 12 hours). With a 12 hour notice, the system effectively manages the unexpected signals with only a 0.57% cost increase compared to the Base Case. However, it struggles with shorter notice times. The 2 hour notice time leads to a 35% peak cost increase, and the 0 hour notice time results in only 49% of throttled energy being reallocated. Implementing V2G functionality further constrains the system, with the 12 hour notice time being the only successful scenario, showing a 3.2% total cost increase at maximum power delivery to the grid. Additionally, an existing IBDRP offering economic incentives through demand charge cost reduction is tested. The 12 hour notice time scenario achieves a 1.8% reduction in operational costs, while the shorter notice times fail to meet charging demands.

The main findings indicate that an IBDRP can be of great value to the office building owner and the DSO. With sufficient notice time, the office building can meet the DSOs throttling demand, and potentially reduce operational costs while also covering the EV owners charging needs.

Sammendrag

Selv om den raske økningen i bruken av elektriske biler (EVs) gir mange miljømessige og økonomiske fordeler, medfører det også nye utfordringer for kraftsystemet. For å gjøre EVs til en del av løsningen og ikke en del av problemet, undersøkes en fleksibel EV-parkeringsplass. Driften av EV parkeringsplassen optimaliseres for å redusere driftskostnadene og toppforbruket til det tilkoblede kontorbygget, samtidig som det tilbys nettjenester gjennom et incentivbasert etterspørsels-responsprogram (IBDRP). Data fra et eksisterende kontorbygg og EV-parkeringsplass brukes sammen med prisdata fra day-ahead markedet.

For å håndtere kortsiktig planlegging i driften av kontorbygget og den tilkoblede EV-parkeringsplassen, implementeres et Receding Horizon (RH)-rammeverk. RH-rammeverket kan håndtere IBDRP-aktiveringssignaler som er gitt med variende varslingstider før parkeringsplassen slår av ladingen. Dekomponeringsmetoden Stochastic Dual Dynamic Programming (SDDP) brukes for å gi fremtidige kostnadskurver til RH-planleggingen. De fremtidige kostnadskurvene implementeres for å strategisk hensynta de langsiktige konsekvensene av å øke månedlig toppeffekt på grunn av effekttariffen. Den kombinerte modellen testes for varierende varslingstider, varierende grader av EV-struping ved IBDRP-aktivering, og en implementering av Vehicle-to-grid-teknologi (V2G) som kan levere strøm tilbake til kontorbygget og/eller nettet.

Modellen evalueres ved bruk av ulike varslingstider (0, 2 og 12 timer). Med 12 timers varsling håndterer systemet effektivt de uventede signalene med en 0,57% kostnadsøkning sammenlignet med modellen uten struping. Derimot så sliter modellen med de kortere varslingstidene. 2 timers varsling fører til en 35% økning i effekt-tariffkostnader, og 0 timers varsling resulterer i at bare 49% av den reduserte energien blir omfordelt. Implementering av V2G-funksjonalitet begrenser systemet ytterligere, med 12 timers varsling som det eneste vellykkede scenariet, som viser til en 3,2% total kostnadsøkning ved maksimal effektlevering til nettet. I tillegg testes et eksisterende IBDRP som tilbyr økonomiske incentiver gjennom reduksjon av effekt-tariffkostnadene. 12 timers varsling gir en 1,8% reduksjon i driftskostnader, mens kortere varslingstider ikke klarer å møte ladebehovene.

Hovedfunnene indikerer at et IBDRP kan være av stor verdi for eieren av kontorbygningen samt for DSOen. Med tilstrekkelig varslingsperiode kan kontorbygningen møte DSOens krav om reduksjon av forbruk, og potensielt redusere driftskostnadene samtidig som de dekker behovene for EV lading.

Preface

This master's thesis was written in the spring of 2024 at the Department of Electric Power Engineering at the Norwegian University of Science and Technology (NTNU). The work done presents a method of optimizing the cost of electricity for a system consisting of an office building and an electric vehicle parking lot, considering an incentive-based demand response program. This master's thesis has been interesting and educational and has also resulted in an paper which is to be presented and published at IEEE EEM 2024 in Istanbul. The paper is attached as a separate attachment along with the master's thesis.

We want to thank our supervisor Kasper Emil Thorvalsen who has been a great support and a great sparring partner throughout this project. This project would not have been the same without his knowledge in optimization techniques and flexible resources. This work is connected to FME CINELDI - Centre for Intelligent Electricity Distribution (Centre for Environment-friendly Energy Research, 257626/E20). We want to thank the Research Council of Norway and the CINELDI partners for financial support for our expenses related to the presentation of the article. Mariona Zhuri from Lede AS and the partners in the Strømfleks project [1] have provided us with load data, which we are very grateful for. We would also like to address our gratitude to NTNU.

Lastly, we would like to thank our fellow students and our parents, for their support during this project.

Andreas Fosse Hansen and Jonas Henrik Pinderud
Trondheim June 2024

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Abbreviations

- **BC:** Base Case
- **BESS:** Battery Energy Storage System
- **C1:** Case 1
- **C2:** Case 2
- **C3:** Case 3
- **CH:** Control Horizon
- **DCC:** Demand Charge Cost
- **DCCR:** Demand Charge Cost Reduction
- **DP:** Dynamic Programming
- **DR:** Demand Response
- **DRP:** Demand Response Program
- **DSM:** Demand Side Management
- **DSO:** Distribution System Operator
- **DSOs:** Distribution System Operators
- **ENS:** Energy Not Served
- **ESS:** Energy Storage System
- **EV:** Electric Vehicle
- **EVPL:** Electric Vehicle Parking Lot
- **IBDR:** Incentive-Based Demand Response
- **IBDRP:** Incentive-Based Demand Response Program

- **kW:** Kilowatt
- **kWh:** Kilowatthour
- **LP:** Linear Programming
- **MPC:** Model Predictive Control
- **MSSP:** Multistage Stochastic Programming
- **NTNU:** Norwegian University of Science and Technology
- **PBDR:** Price-Based Demand Response
- **PBDRP:** Price-Based Demand Response Program
- **PH:** Prediction Horizon
- **RH:** Receding Horizon
- **ROH:** Rolling Horizon
- **SDDP:** Stochastic Dual Dynamic Programming
- **SDP:** Stochastic Dynamic Programming
- **SOC:** State of Charge
- **SP:** Stochastic Programming
- **TSO:** Transmission System Operator
- **V:** Volt
- **VAT:** Value Added Tax
- **V2G:** Vehicle to Grid
- **V2G-BC:** Vehicle to Grid-Base Case
- **V2H:** Vehicle to Home

Chapter 1

Introduction

1.1 Motivation

The global transition towards sustainable energy and transportation has seen a rapid increase in the adoption of Electric Vehicles (EV). While this shift brings forth many environmental and economic benefits, it also presents new challenges for the power systems. As EVs have a high energy consumption when charging, several studies have predicted the consequences of the rapid integration of EVs. One of these studies [2] analyses how the increase in EVs has strained local distributional grids in Norway. The study found that EVs have increased the total energy electricity use in Norway by 0.42% in 2020. Statnett also states that the electric energy use for transportation will be 20 TWh by 2030 in Norway [3]. From this, it is evident that the transportation sector is going to make a substantial impact on the grid.

One of the main challenges of the rapid increase of EVs is the increased peak loads in the grid. A solution to mitigate peak consumption for the Distribution System Operators (DSOs) is the integration of capacity-based grid tariffs. For instance, the monthly demand charge puts a cost on on the highest peak single-hour import in the month, which has been present at commercial user-level before, but not at household-level [4]. From this, the EV charging operators are motivated to optimize their energy use over a monthly horizon in combination with the day-to-day operation.

Office buildings and other commercial premises experience changes in their demand profiles due to the integration of EV charging facilities, affecting peak import accordingly [5]. DSOs can use Incentive-Based Demand Response Programs (IBDRP) as an incentive to reduce grid problems. An example is Glitre Nett which offers a Demand Charge Cost reduction (DCCR) as an incentive [6]. This reduction is based on the notice time before activation, where lower notice times will result in a higher reduction. By strategically utilizing the EV Parking Lot (EVPL) as a flexible resource, grid services can be provided to the DSOs as demand response (DR) to reduce operational costs for service provider.

1.2 Objective/Contributions

This thesis aims to evaluate the costs and potential benefits of a flexible EVPL connected to an office building participating in an IBDRP. These programs offer incentives for modifying consumption patterns. This study focuses on optimizing EV charging and minimizing the associated costs, and examines how participation in the IBDRP affects operational performance and costs. Additionally, the thesis explores Vehicle-to-Grid (V2G) technology to analyze how the technology affects operational performance, with and without IBDR-participation. The IBDRP allows participation with varying notice times before activation, impacting operational decisions due to the uncertain activation notification. To the author's knowledge, limited work has looked into IBDR programs with different notice times. In this thesis this is addressed by integrating notice time into scheduling decisions within the Receding Horizon (RH)-framework to react to the flexibility needs.

The main contributions of this work are the following:

- A RH-framework is developed to operate an office building with a flexible EVPL, subject to hourly spot prices and monthly demand charge.
- An SDDP-framework is applied to account for long-term impacts of raising the peak due to the monthly capacity-based Demand Charge Cost (DCC)
- The RH-framework is investigated for IBDR-participation, where load reduction is requested by the local DSO. With a variation in notice times before activation and different quantities of throttled power, the EVPL scheduling is analyzed to provide insights on consequence of IBDR for the office building.
- This work investigates the operational impact with V2G available, where the EVPL scheduling is analyzed to provide insights on the consequence of a throttling activation which also requires supplying energy back to the grid.
- Analyze the consequence of an IBDRP when economic incentives are provided which reduces the DCC based on the notice time of the IBDR-signal.

1.2.1 Limitations

The model does not consider any dynamic limitations in power flow that might occur in a real-life system. No investment costs are considered, only the operational costs associated with energy consumption. The modeling of the EVPL is based on historical data on an aggregated level, which limits the data available on the individual EVs.

1.2.2 Structure

The rest of this thesis is structured as follows:

Chapter 2 Theory: Provides theory about relevant topics for this thesis. Overall, this chapter will consider the electricity grid, the Norwegian pricing model, Demand Side Management (DSM), and EVs.

Chapter 3 Literature review: Provides literature review on previous research of EVs and DRP.

Chapter 4 Optimization theory and literature: Provides theory about general optimization theory and more specific theories for the two optimization methods used in this thesis. It also provides a literature review on optimization models that have been used for similar purposes as this thesis.

Chapter 5 Method: Presents the general procedure for the analysis of this thesis. This chapter includes how the system is modeled, including the office building and the EVPL. It also presents the frameworks for the SDDP and RH algorithms

Chapter 6 Case study: Presents the real-life office building with a flexible EVPL, and the spot prices for the month that is going to be analyzed. Relevant input data to the model is presented. Lastly, this chapter presents the different cases that are to be analyzed.

Chapter 7 Results and discussion: Presents the results coupled with an analysis and discussion of the results. The discussion focuses on the economic and operational performance, and emphasize how the peak changes and the energy shifts due to sudden IBDR-signals.

Chapter 8 Conclusion: Concludes and summarizes the key findings of this study, along with suggestions for further work.

1.2.3 List of Publications

During this Master's thesis, a paper was developed based on parts of the research. The paper is to be presented and published at IEEE EEM 2024.

> **Paper I:** J. H. Pinderud and A. F. Hansen. "Operation of an EV parking lot subject to capacity-based grid tariffs offering demand response", to be presented and published at: *IEEE EEM 2024*

Chapter 2

Theory

In this chapter, a theoretical framework is presented so that the motivation and methodology in the thesis can be better understood. First, an overview of the electricity grid and power markets is presented with the different levels and components. Afterward, a section is presented on the ways of incentivizing flexibility through demand response programs. Lastly, theory is presented on the EV, EV charging, and associated technologies.

Some of the content in Section 2.1.1, 2.1.2, 2.2.1, 2.2.2, 2.3, 2.3.1, 2.3.1.1, 2.3.1.2, 2.4.1 and 2.4.1.1 was originally part of the preceding project thesis as presented in [7]. The content has been reviewed and modified in accordance with the objectives of this Master's thesis.

2.1 Electricity grid

An electricity grid is defined as a network where electricity is delivered from producers to consumers. The main elements in an electricity grid are power stations, electrical substations, transformers, transmission systems, and distribution systems [8]. In this section, the transmission- and distribution system is further explained.

2.1.1 Transmission System

The Transmission System Operator (TSO) is the administrative body that is responsible for building, managing, and maintaining the transmission grid [9]. The transmission grid connects producers with consumers in a nationwide system together with interconnections between other countries [10]. The transmission grid carries a high voltage, usually 300 to 420 kV, with certain parts of Norway having lines carrying 132 kV. Due to the cost of establishing and maintaining such a large infrastructure, a TSO is often a monopoly and therefore subject to strict regulations. In day-to-day operations, a TSO is responsible for maintaining a continuous

balance between electricity supply from power stations and the demand from consumers, and also ensuring the provision of reserves that will allow for sudden contingencies [11]. The only TSO in Norway is Statnett, which is a state-owned company under the oil- and energy department.

2.1.2 Distribution System

The Distribution System Operator (DSO) is a key entity in the energy sector responsible for managing and operating the distribution system, which delivers electricity from the high-voltage transmission system to end-users. DSOs are responsible for the real-time operation of the distribution grid, ensuring the balance between electricity supply and demand. DSOs perform maintenance on the grid, including power lines, transformers, and substations. The service provided ensures a reliable power supply to end-users, by working to make the grid resilient against disruptions. In Norway, the distribution grid is divided into two levels [10]: The **regional distribution grid**, serves as the intermediary, connecting the transmission grid to the distribution grid. This grid can also include higher voltage production and consumption radials. The regional typically operates at a voltage of 33 to 132 kV. The **local distribution grid** supplied electricity to smaller end-users. This grid operates at voltages up to 22 kV and is split into high-voltage and low-voltage segments. The threshold between these segments is 1 kV, with the low voltage distribution to standard consumers normally carrying 400 V or 230 V.

2.1.3 Challenges in the Norwegian Power System

Increased renewable power production and the electrification of the transport sector are necessary to meet global emission goals but also bring forth many challenges for the grid operators. New larger renewable power production such as solar and wind is connected to the grid through converters [12]. This can worsen the grid stability, especially in Norway where the grid is built around hydropower-generation which is connected to the grid through synchronous generators. The synchronous generators are synchronized to the grid frequency while strengthening the frequency stability of the system. There is also an increase in distributed energy production, mostly through solar, which can cause local grid-stability issues [13]. The distributed energy production is more often located in weaker parts of the grid further out in the system, which is more prone to grid issues. Problems which were pointed out due to local solar production are voltage quality issues and thermal loads in cables and transformers.

The TSO in Norway, Statnett, is already experiencing situations where they do not have enough grid-stability reserves [12]. Grid-stability reserves are flexibility services that are provided and activated when there is a need to stabilize the grid or frequency. With increased variable renewable energy production, the need for reserves is increasing. Not having enough reserves can lead to market constraints, and in worst-case a forced disconnection of consumers or producers. The increasing need for reserves leads to economic possibilities for both existing and new market operators. Statnett states that *consumption and energy-storage has to*

contribute more than what we are used to. This encompasses flexible use of electricity for heating in buildings and industrial applications, charging of vehicles and the production of green hydrogen [12]".

2.2 The Norwegian Pricing Model

2.2.1 The power market

In Norway, all power trading is done through the European power exchange Nord Pool [14]. Nord Pool was founded in 1993 as Statnett Marked AS, as the Norwegian government deregulated the market for power trading. Throughout the years, Nord Pool has expanded further out in Europe, and now provides services in most of northern Europe. Nord Pool is an exchange built up from two main markets, namely the financial market and the physical market. Based on the scope of this thesis, only the physical is further discussed.

As presented in Figure 2.2.1, the physical market includes the day-ahead and intra-day market [14], while the TSO runs the Capacity market [15].

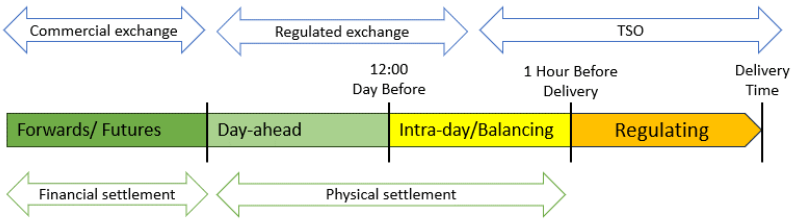


Figure 2.2.1: The Nord Pool Market [16]

2.2.2 The Day-Ahead Market

In the day-ahead market customers can sell or buy energy for the next 24 hours in a closed auction. Orders are matched to maximize social welfare while taking network constraints provided by transmission system operators into consideration [17]. This ensures that the power system is balanced before delivery [18]. Available capacities on interconnectors and in the grid are published at 10:00 CET (Central European Time). After this, the buyers and sellers have until 12:00 CET to submit their final bids to Nord Pool auction for delivery hours the next day. The market results are typically published before 13:00 CET and are referred to as spot prices [17]. Between the clearing of the spot price market and the delivery hour, the market participants can adjust their market position in the intra-day market [18].

2.2.3 Grid tariff

To cover the costs of grid operation, DSOs typically implement a grid tariff, which requires end-users to pay for the transportation of energy. The grid tariff usually consists of a volumetric cost based on the energy consumption, a fixed cost, and on commercial buildings there is a capacity-based cost which in this thesis is called the demand charge. Residential customers were introduced to the capacity-based grid tariff in 2022. The rates on the grid tariff and the contents are set by the respective DSOs. The Norwegian Energy Regulatory Authority (RME) overlooks the income from the grid tariffs and checks that the DSOs income does not surpass the regulated amount [4].

The grid tariff can differ between commercial customers and residential customers, as presented by the DSO Lede in [19]. Commercial customers with a yearly consumption of less than 100,000 kWh are usually on similar terms as residential customers. This includes a monthly demand charge based on the average of the three peak hours in a month. A monthly fixed cost is included which covers a fee to the energy fund, also called the ENOVA-fee. The electricity fee is included on all power consumption, which is a volumetric cost that differs between the summer and winter months. VAT on electricity is set at 25%.

For larger commercial customers, the grid tariff is structured similarly to the smaller commercial customers, with a fixed cost, an electricity fee and VAT. The difference is that the DCC usually only considers the single highest peak hour of the month, instead of averaging the three highest hours. The demand charge can also be seasonal, being higher in the months when the grid is more constrained. Lastly, a seasonal volumetric charge on reactive power consumption can be included, usually in NOK/kVAR/month.

2.3 Demand Side Management

In this master's thesis, the definition of DSM by Peter Warren is used; *Demand-side management (DSM) refers to technologies, actions and programmes on the demand-side of energy metres that seek to manage or decrease energy consumption, in order to reduce total energy system expenditures or contribute to the achievement of policy objectives such as emissions reduction or balancing supply and demand.* [20]

DSM is a method that enables the energy sector to influence and manage electricity consumption patterns on the consumer side of the market. This is to encourage consumers to reduce their demand during peak hours or shift their energy use to off-peak periods. Three of the most common ways to for consumers to purposely alter their demand patterns are through peak clipping, valley filling, and load shifting [21]. In Figure 2.3.1, different DSM techniques are presented and visualized.

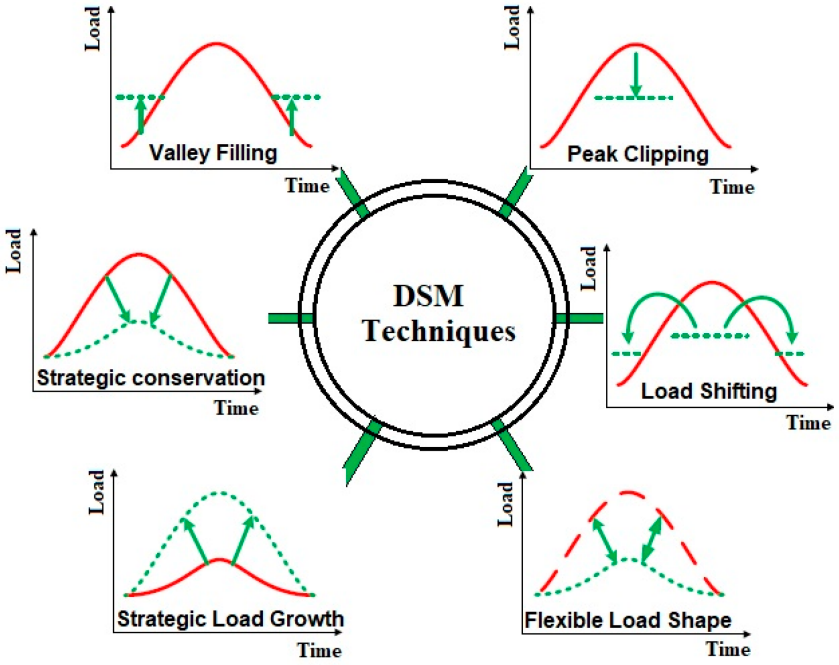


Figure 2.3.1: Illustration of how different DSM techniques can be used to lower a peak. Figure from [21]

2.3.1 Demand response

A common initiative of DSM is Demand Response (DR). DR can be defined as the changes in electricity usage by end-use customers from their normal consumption patterns in response to changes in the price of electricity over time or grid reliability concerns [22] [23]. Demand Response Programs (DRP) can be used to reduce energy costs, reduce the variation in electricity prices, and make a more reliable power system. The DRPs performance can be measured by peak load reduction and demand elasticity [24]. It can be particularly challenging to create demand charge programs which fairly reflects the costs to the DSOs [25]. In Norway, a DRP was implemented for all household end-users in 2022 when a demand charge was implemented in the grid tariff, with the intention of making the customers aware of their peak power usage [26].

DR programs are particularly relevant in the context of EVPL due to the amount of electricity that can be consumed when multiple EVs are charging simultaneously. In this research, the two most common categories of DRPs are discussed, namely price- and incentive-based demand response.

2.3.1.1 Price-based demand response

Price-Based Demand Response (PBDR) is a strategy that encourages consumers to adjust their electricity consumption patterns in response to price signals [27]. The purpose is to signal

the customers to change their consumption to better reflect the grid situation. The customers are "punished" with higher electricity costs if the signal is ignored. A common way of introducing PBDR is by providing accurate spot price data. Before the introduction of smart meters in Norway, the spot price data was given as a monthly average. By 2019, all Norwegian homes had a digital smart-meter installed [28]. The recent introduction of smart meters in Norwegian households has made hourly power consumption data available for power and grid companies, and spot price data is now provided in an hourly resolution. Hourly spot prices make it possible for the end-users to adjust their demand more accurately to save costs and help reduce peaks in the grid.

Another well-integrated PBDR program is the demand charge, which was already mentioned in Section 2.2.3 as being a part of the Grid Tariff. Demand charge uses the consumers peak power to charge the consumers with the intention of reducing power peak [29]. The demand charge forces the short-term planning of electricity consumption to consider the monthly implications of raising the peak, complicating the optimization of energy use. There is varying practice when determining the basis for the demand charge. While some use the customer's maximum power consumption per month, others base it on the average power consumption from multiple measurements over the same period. However, both approaches are considered to be within the current regulatory framework [4].

2.3.1.2 Incentive-based demand response

Incentive-based Demand Response (IBDR) provides fixed or time-varying incentives to participating customers for reducing their electricity usage [30]. IBDR works by sending signals to the end-users, notifying them to reduce their electricity consumption in periods when electricity demand is high or when there is a need to stabilize the grid. This might involve adjusting the heating, delaying the use of certain appliances, or adjusting the EV charging. IBDR being a voluntary agreement, consumer engagement and willingness to participate are important factors for IBDR being a successful strategy.

IBDRPs have been utilized throughout energy grids which need higher reliability in peak demand hours. A example is Tibbers smart charging [31], where the customers EV is automatically charged when the electricity price is the lowest. Here, a 20% lower charging cost is guaranteed, where Tibber in return can use the EV charging to help balance the grid. In California, a demand response auction mechanic was introduced to the energy market in 2014, which is a pay-as-bid pilot in which investors-owned utilities seek monthly demand response system capacity [32]. Further, in Norway, a *Disconnectable Tariff* has been offered by grid companies for some time. An example is Glitre Nett and their electricity tariff for large businesses and commercial customers with adequate power consumption [33]. Here a percentage-wise reduction in the DCC is offered for consumers, in which the size of the reduction is based on how preemptive the signal for the request must be sent.

2.4 Electric Vehicle

There has been a rapid increase in the number of EVs in Norway in recent years. The percentage of EVs in the Norwegian car park was 24% in July 2023 [34]. Further, globally Norway has the most EVs per capita and the most public chargers per EV. In 2022, 58% reported that they used a car to travel to and from work as the driver, and 5% reported that they used a car to travel to work as a passenger [35]. With this in mind, the EVs spend a significant amount of time parked at the office parking lot. All new buildings must facilitate EV charging after a legal regulation in 2021 [36]. As EV charging can strain the local grids, further explored in Section 3.1.2, having intelligent charging such as smart charging to distribute the loads can be important in the future.

2.4.1 EV charging

In this section, the goal is to map the categories of chargers in the market, and how they can facilitate EVs a flexible resource. An essential distinction among EV chargers lies in their classification as either passive or smart chargers. A passive charger is defined as a charger that supplies a constant power flow, without any consideration to the battery status [37]. A smart charger can consider the battery status, resulting in a dynamic power flow. A smart charger can also enable intelligent charging management when connected to the grid, using a communication module [37]. Enabling communication with the grid, a smart charger can charge an EV during off-peak hours. The integration of smart chargers can also contribute to the efficiency, reliability, and sustainability of the electric grid [37][38].

2.4.1.1 Categorization of EV chargers and their function

In Norway, the charging standard is divided into four different modes, which the International Electrotechnical Commission presents under the standard IEC 62196 [39].

Mode 1 chargers refer to regular or slow charging from standard power outlets [39], and are defined as passive chargers. These chargers are the least safe chargers, as they are connected directly to the grid with no built-in safety functions. In Norway, this includes charging on a 230V 1-phase socket and a 400V 3-phase socket with a charging current of up to 16A. Modern EVs no longer use this type of charger, and the maximum power output of the Mode 1 charger is 2,3 kW [40]. The use of this charger would only have a flexible potential by manually disconnecting the charger.

Mode 2 chargers are also defined as passive chargers, charging from standard power outlets, but the charging cable is semi-active [39]. The cable has built-in safety features that can partially handle potential risks that may occur during charging. In Norway, this includes charging on a 230V 1-phase socket and a 400V 3-phase socket with a charging current of up to 32A. The maximum power, similar to Mode 1, is also 2,3 kW [40]. Mode 2 chargers flexibility

potential is also restricted by the manual disconnection of the charger. However, a slow charger like this will have a small effect on the capacity of the grid.

Mode 3 chargers are defined both as slow chargers and fast chargers [39]. Mode 3 chargers are smart chargers, making them able to postpone the charging until the time of day when electricity is cheapest. The control and safety functions are integrated into one charging point, which is known as a charging station. Mode 3 chargers can also enable smart home solutions, by increasing or decreasing the charging current depending on the overall power consumption in the house, making the EV act like a flexible resource. The charging power for Mode 3 varies from 7,4 - 22 kW, based on the electricity grid [40]. Flexibility can be provided to the DSO by the Mode 3 charger, especially if several Mode 3 chargers are managed simultaneously to reduce the charging load when grid demand is high.

Mode 4 is defined as direct current fast charging [39], and is also considered a smart charger. The charger is integrated into the charging station, which has a rectifier that provides DC directly to the battery. In 2019, the most common power-level for Mode 4 charging stations was 50 kW, but this is constantly increasing as more EVs are being designed to have the opportunity to be charged with higher power [40]. Mode 4 chargers can be a challenge to the grid because of the high charging power. However, Mode 4 chargers can enable that EVs participation in the balancing market. As the charging stations can draw high amounts of power, a throttling signal from the DSO or aggregators can be of great value for grid stability.

2.4.1.2 Bi-directional charging

A bi-directional charger is a charger with a dual-direction power flow, which means that the power can be transferred to and from the EV [41]. The vehicle-to-building (V2B) technology makes it possible for the EV to extract energy from a building, or transfer its energy to a building. Thus, V2B technologies can contribute to the development of smart buildings [42]. Vehicle-to-grid (V2G) refers to the bi-directional power flow between EV and the power grid [43]. V2G technology can be used to feed power back into the power grid during peak hours, resulting in grid balancing and peak shaving.

V2G-enabled chargers also enables smart charging, which means that they charge the EVs during low-demand periods. The batteries in the EVs can store energy during times of high production and low demand, and discharge during times of low production and high demand, acting as a Battery Energy Storage System (BESS) [43]. Continuous charging and discharging can affect an EV batteries lifespan [43]. Participating in an IBDRP can give economic incentives and motivation to the EV owners and EV charging station owners to integrate the V2G technology. Today, only a few EVs are capable of bi-directional charging [44], but more EVs that are entering the market are integrating this feature [45]. Bi-directional charges can cost more due to the additional components needed to manage the bi-directional power flow. Since both V2G and V2B technologies employ bi-directional chargers, in this study, V2G from here on encompasses both the bi-directional power exchange between the EVPL and the power grid, as well as between the EVPL and building.

Chapter 3

Literature Review

This chapter presents the relevant framework and literature on concepts such as EV charging and flexibility, DR, use of flexibility in IBDRPs, and model approaches. The work in this chapter is used to define the scope of this thesis, highlighting what work is already done on the different topics, and identifying the topics that needs to be further addressed.

Some of the content in Section 3.1.3, 3.1.4, and 3.2.1 were originally part of the preceding project thesis as presented in [7]. The content has been reviewed and modified in accordance with the objectives of this Master's thesis.

3.1 EV Charging and Flexibility

In this chapter of the literature review, the focus is on the consequences of EV integration on the grid, charging strategies, and V2G technologies. This is done to find out how the literature extracts flexibility out of EVs, and how flexible charging and inflexible charging can be a problem for the grid.

3.1.1 Results of increasing EV diffusion

With a goal of all new consumer cars sold in Norway being zero emission from 2025 [46], either hydrogen or electric, the diffusion of EVs and the impact on the energy system has to be investigated. Henriksen et al. [2] studied the impact of increasing numbers of EVs in Norway. They find that the diffusion of EVs in Norway has significantly strained local electricity grids, and the increase in total electricity use because of EVs is calculated to be 0.42% of the total electricity use in Norway as of 2020. With electric energy use for transportation in Norway forecasted to be at 20 TWh by 2030 [3], transportation is going to make a substantial impact on the grid.

A study done by P. Wangsness and A. Halse [47] investigated the impact of EV density on local grid costs in Norway based on a historical dataset spanning from 2008 to 2017. It is stated that the diffusion of EVs within a DSOs operational area is often associated with higher local grid costs. It is also found that an increase in EV diffusion in areas where the DSO has a higher density of customers results in a lower effect on the local grid cost.

In an article written by Affonso and Kezunovic [48] it was found that the installation of EV chargers in a commercial building increased the daily energy consumption by 23%. This increase in electricity use creates issues in neighborhoods, overwhelming the old grids. Stringent regulations and safety concerns have led to a shift from simply plugging EVs into a regular socket, to installing dedicated charging stations, contributing to a substantial market for various EV chargers in Norway [2]. This shift aggravates the strain on local power capacity, leading to congestion being more of a concern. This necessitates costly upgrades to low-voltage networks as DSOs struggle with increasing demand for EV charging infrastructure. However, newer and faster charging stations provide not only a significant increase in safety but also enable EV charging which considers peak loads which in turn can help with grid congestion.

3.1.2 Grid problems due to EV integration

The intense integration of EVs has a huge energy requirement and has serious effects on the energy sector [49], such as overloading, voltage instability, voltage drops, voltage unbalance, energy losses, harmonics, frequency variation, and consequently reductions in supply quality [50].

A study from Canada considered the integration of EVs into the distribution grid, where the impacts of uncoordinated charging were evaluated on a real test feeder [51]. It was found that for normal and fast charging scenarios, the uncoordinated charging led to unacceptable voltage drops in remote buses, and the feeder line loading constraint was violated at 15% and 25% EV penetration for fast charging and normal charging respectively. Another case study was done on the power system on the Danish island of Bornholm [52], which analyzed the impacts of EV penetration on the grid. For the uncontrolled study, the feasible integration of EVs was 10%, while the controlled charging could handle the integration more effectively. Voltage drops are found to be the limiting factor, and the peak demand in the network is increased by 31% for a 50% EV integration controlled charging scenario. From the literature, it is evident that increased EV penetration with uncontrolled charging significantly increases peak demand and strains the grid. Having more flexible EV charging is one way to combat this.

3.1.3 Flexibility from EV charging

Knezovic et al. [53] state in their research that *DSOs have historically dealt with grid-security challenges by making investments in grid reinforcement. However, an alternative solution, enabled by the rollout of smart meters and high penetration of flexible loads, would be the increased use of flexibility services. Flexible loads, with EVs at their forefront, can modulate*

their consumption or even inject power back into the grid, depending on current grid conditions.

As the battery system in EVs can be modeled as a flexible load, looking into BESS literature can also be relevant. In an article written by Y. Hu et al in 2022, they investigate the potential utilization of BESS in some of the major European electricity markets [54]. It was found that BESS can be a feasible solution in some parts of Europe, but has more limited potential in Norway. They state that this is due to the Norwegian high penetration level of hydropower. However, the study shows that the potentially profitable utilization increases every year in Norway. The need for power increases and hydropower in Norway is already well developed, making BESS a potential part of the solution for the future grid in Norway.

Winschermann et al. studied the significance of historical data in enhancing EV charging strategies at office buildings [38]. Given the challenges posed by the high energy demands of EVs on the electrical grid, smart charging was proposed as a potential solution. In the study, historical data from an office parking lot, specifically focusing on arrival times, energy requirements, and dwell times, was used to determine how past data mirrors real-time operations. An algorithm was used to construct a power charging profile for EVs based on user-specific parameters. Using a greedy charging approach in office settings can result in high power peaks due to synchronized arrivals. Here, greedy charging entails charging without an energy management system, which charges at the maximum available power from the plug-in. Using historical data for efficient planning helped reduce the peaks by 50%. Complete historical data which concerned arrival and departure times and energy requirements, was found to be essential for efficient planning. In an article written by Affonso and Kezunovic [48], smart EV charging in an EVPL for a commercial building is used to reduce the cost of operation for the building owner while also reducing the strain on the local grid transformer. Without the smart EV charger, the transformer was overloaded in the summer season, which led to accelerated aging of the transformer.

Understanding the motivation for the end-users to participate in smart charging and the flexible use of their EVs is important. The study by I. Henriksen et al. [2] presents the key motivations for the adoption of smart chargers for EVs. Participants emphasized the safety and faster charging speed of smart chargers compared to regular sockets. Many of the participants were interested in smart home technology, which in turn made smart charging very interesting. Additionally, the recognition of practical and economic benefits, such as improved load management and potential incentives from the pilot project, played a significant role in the decision-making process. The flexibility of smart charging, allowing users to delay charging during periods of lower electricity demand, was also seen as advantageous. In conclusion, the end-users adoption of smart chargers was influenced by a combination of safety considerations, technological interest, practical and economic advantages, and the flexibility offered in the charging process [2].

3.1.4 Vehicle to grid

When optimizing the flexible use of an EVPL, V2G is a technology that could help provide the wanted flexibility. There are three main domains for V2G studies, namely the transmission grid, distribution network, and consumer [41]. This literature study focuses on the consumer domain.

A study conducted by Yee Ting Chai et al. [41] introduces a two-stage optimization method for EVs in a V2G program at an office building. For the V2G program to work, the scheme must address both the needs of EVs and building owners. For building owners, EVs need to be reliably present to supply power. EV owners, on the other hand, should benefit financially without V2G affecting their travel plans. To achieve this, the EV owners are allowed to reach their desired State of Charge (SOC) at any time by raising the energy anxiety factor. The proposed method uses day-ahead optimization based on expected loads and EV patterns, aiming to cut electricity costs. Then, in the second stage, a real-time optimization adjusts to actual EV behaviors. The simulations show promising results, although it is shown to be difficult to consider the monthly peak demand when optimizing for shorter intervals.

Jingli Guo et al. [55] explores the impact of V2G services on EV battery degradation. The study introduces EVs as a solution to environmental issues and highlights the potential of V2G to support electricity grids. They also express concerns about the possible acceleration of battery degradation due to frequent charging and discharging cycles associated with V2G. Findings from different studies vary on whether V2G has acceptable or detrimental impacts on battery degradation. Factors affecting battery degradation include battery chemistry, charging/discharging regimes, types of V2G services, and service frequency [55]. Appropriate V2G management, such as optimizing cycling during idle states, could minimize battery degradation. It is concluded by emphasizing the need for accurate battery degradation models and improvements in battery management for V2G deployment. Zhoa [56] states that aggregators should compensate the EV owners for battery degradation when providing V2G services to the power grid.

3.2 Demand Response

As mentioned in Section 2.3.1, DR is a method where the objective is to manage and modify electricity consumption in response to external signals, such as changes in electricity prices or grid reliability concerns [23]. The goal is to research how the two of the most common classifications of DRPs, PBDRP and IBDRP, are implemented and used.

3.2.1 Demand charge

The demand charge is categorized as PBDR and is a part of the electricity bill for end-users. It is based on the highest level or levels of electricity demand during a specific period, typically

measured in kilowatts (kW) [29]. In this section, the implementation of the DCC is investigated together with the impact of considering the long-term impact of short-term decisions in optimization models.

In the study done by Y.He et al. [57], the DCC in a charging station for an electric bus system was analyzed. DCC is defined as a part of the total system cost, and the text illustrates how both the total system cost and demand charge increase with an increase in the demand charge rate. When including an energy storage system, the total cost for the system was reduced by 9.2%. In the study by K. Thorvalsen et al. [58], the implications of the new capacity-based grid tariffs in Norway, which include a demand charge, are investigated. Stochastic Dynamic Programming (SDP) is implemented to represent the long-term impacts of the demand charge. The method was applied to a real Norwegian household, to strategically determine the cost-optimal peak power and the expected total electricity cost. Two scenarios were investigated: one where the future cost from the grid tariff, i.e. the demand charge, is unknown and not considered, and another with perfect information about stochastic inputs. The results revealed a 36.1% increase in cost when not considering the demand charge.

Operating an energy system with a monthly demand charge can pose challenges when trying to identify the optimal peak-import level to aim for. A metaheuristic approach was conducted in [59] to balance the short-term cost of operation for a household with the long-term impacts of the demand charge, using a user-defined weighting constant. A high value for the weighting constant indicated that the energy management system weighted the real-time price less. A low weighting constant would indicate that the energy management system pays more attention to the real-time price, resulting in a high DCC. When the the weighting constant was optimal, the total cost was reduced by 64,76 % compared to when the system did not have a management system.

3.2.2 Use of flexibility in IBDRPs

In this section the goal is to showcase how earlier studies used IBDRPs to provide the DSOs with the needed flexibility. The studies have also looked into how and what the incentives should be for participation. Latly, this section presents a study that investigates the consequences of different notice times prior to an IBDR-signal.

Rana et al. studied how a residential community could provide ancillary services like voltage support and loss minimization to a DSO [60], assuming that the ancillary service provider has information on the status of the grid. It was found that when the objective was to minimize the losses, the active losses were reduced by 4.2%. In the article written by Berg et al. [61], the primary objective was to study how an energy community and a DSO can coordinate to improve the voltage profile in the grid. A minimum compensation of 15 € per year to the energy community for the voltage services was found, based on a yearly cost increase of 0.12% because of increased electricity use and battery degradation. It is suggested that the energy community and DSO should collaborate to make sure that the battery characteristics are sufficient to improve the voltage profile in the grid.

Zhang et al. [62] studied smart EV charging at non-residential sites to reduce peak consumption using a proposed water-filling algorithm with an IBDRP. The work found that an 80% DCCR of the theoretical maximum could be achieved with the IBDRP, leading to significant savings to the retail venue. In the study conducted by Yee Ting Chai et al. [41], where a V2G program at an office building was implemented, a detailed analysis of the costs and benefits associated with V2G to reduce peak electricity import is included [41]. To encourage EV participation in V2G, an IBDRP is implemented, which rewards the discharging of EVs. This research indicates the potential of V2G in commercial buildings, benefiting both building and EV owners. The studies presented show that the integration of IBDRPs with other flexible resources offer a promising opportunity for enhancing grid stability, reducing peak demand, and optimizing resource utilization when various stakeholders in the energy system collaborate.

Study [63] investigated how different notice times in a DRP, with continuous updates of wind forecasts, affected flexible consumers. The DRP used was demand bidding, which is defined as a market based program. The study found that a long notice time is associated with higher forecast deviations due to the uncertainty in long-term wind power predictions. A upside however, is that the longer time period means the energy demand is more flexible, allowing consumers to adjust their consumption schedules. Shorter notice time resulted in smaller forecast deviations as the certainty of predictions in wind power increased. However, the flexibility of energy demand decreased, which made it harder for consumers to reschedule their consumption at short notice.

Chapter 4

Optimization Theory and Literature Review

This chapter presents the relevant optimization background theory and literature for this thesis. The theoretical background presented is used to establish the method used in this thesis. First, the relevant theory is presented, followed by a representation of how earlier studies have implemented the optimization strategies, and for what purpose.

Optimization Theory

4.1 Deterministic Programming

The field of optimization belongs to the field of applied mathematics and encompasses the use of mathematical models and methods to find the best alternative in decision making situations [64]. An optimization problem is a mathematical approach to a physical problem, where the problem is analyzed to control the output through the input variables. A deterministic optimization problem will always give the same result given the same input. That is, there is no randomness or uncertainty in the modeling.

4.1.1 Linear programming

Linear Programming (LP) is a method of calculating the optimal value of a mathematical model where the objective can be expressed as a linear function that is limited by a set of linear constraints [64]. In equation 4.1, a general representation of an LP problem is given.

$$
\begin{aligned}
\min \quad & z = \sum_{j=1}^{n} c_j x_j \\
\text{s.t.} \quad & \sum_{j=1}^{n} a_{ij} x_j \leq b_i, \quad i = 1, \ldots, m \\
& x_j \geq 0, \quad j = 1, \ldots, n
\end{aligned}
\tag{4.1}
$$

z represents the objective function, x $\forall$ j are the decision variables, while a_{ij} and b_i are parameters representing the information known in advance of the solution process. The optimal decision variables are often denoted x^*, and the optimal objective function value is denoted $z^* = f(x^*)$. An additional requirement for an LP problem is that all variables are continuous, i.e. $x \in R^n$.

4.1.2 The dual problem

Associated with every LP problem, there is a *dual* problem which is defined with exactly the same input data as the original *primal* problem [64]. The dual values can provide insight into a solution and help in sensitivity analyses to changing parameters.

4.2 Stochastic programming

Stochastic Programming (SP) is a methodology used to handle uncertain information in optimization, and making decisions under uncertainty. Randomness is now introduced to the problem for a more realistic approach.

4.2.1 Multistage stochastic programming

Multistage Stochastic Programming (MSSP) is an optimizing programming model that includes multiple stages that for each stage presents new stochastic information. A common example is that the stages represent time periods. If it in reality is practically impossible to perfectly forecast uncertain state variables in the future, the problem is essentially stochastic [65]. In each stage, the optimal solution is determined for a number of subproblems. The decision made in this stage depends on the history up to the current stage. The set of possible future sequences of outcomes in the problem are called scenarios [66]. The scenarios can be ordered in scenario tree, as illustrated in Figure 4.2.1. Illustrated are the nodes n for each realization, the time-stages $t = [1, 2, 3]$ and 4 scenarios ω.

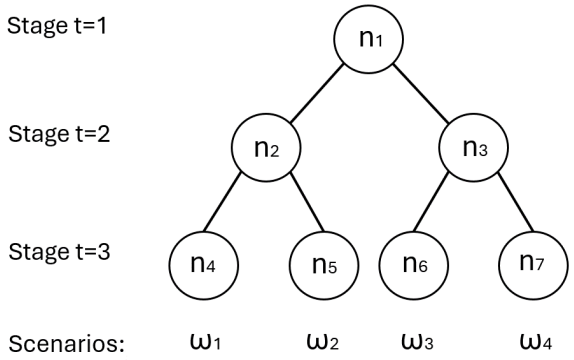


Figure 4.2.1: Illustration of a scenario tree with 3 stages, 4 scenarios, 7 nodes

4.2.2 Decomposition techniques

In MSSP, the problem, and the scenario trees, quickly become very large. State variables are used to describe the state of the system at the current stage. These variables have the Markov property, meaning only the current state matters, not the history leading up to the state [67]. A way of solving these large multistage problems is by using decomposition techniques. One of these decomposition techniques is SDP.

As commonly done with other approaches, SDP can be used to minimize costs. For SDP, the problem is decomposed into several one-stage subproblems, where the objective is to minimize the sum of immediate and future operating costs [68]. The typical approach is to form a backward recursion resulting in an optimal decision associated with each state at each stage [66]. The SDP recursion requires the enumeration of all combinations of initial values [68]. As a consequence, computational efforts increase exponentially with the number of state variables. A problem that can arise for Dynamic Programs with multiple states, as seen in SDP, is that the number of discrete states per stage grows exponentially with the number of state variables, which is also known as *the curse of dimensionality.*

SDDP is an optimization method that can be applied to MSSP problems. No discretization is necessary, and the curse of dimensionality is avoided. *Unlike SDP, where the entire state space is explored, SDDP explores the most interesting states based on sampling from the uncertainty distribution* [69]. A scenario tree for a three-stage problem is shown in Figure 4.2.2. An important observation from the figure is that the number of scenario-states is the same for stage 2 and stage 3. Here, the scenario tree does not fan out as in SDP, shown in Figure 4.2.1.

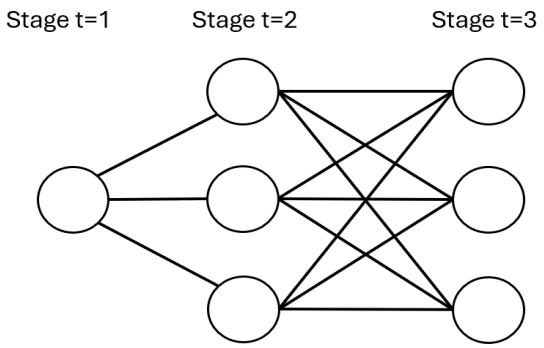
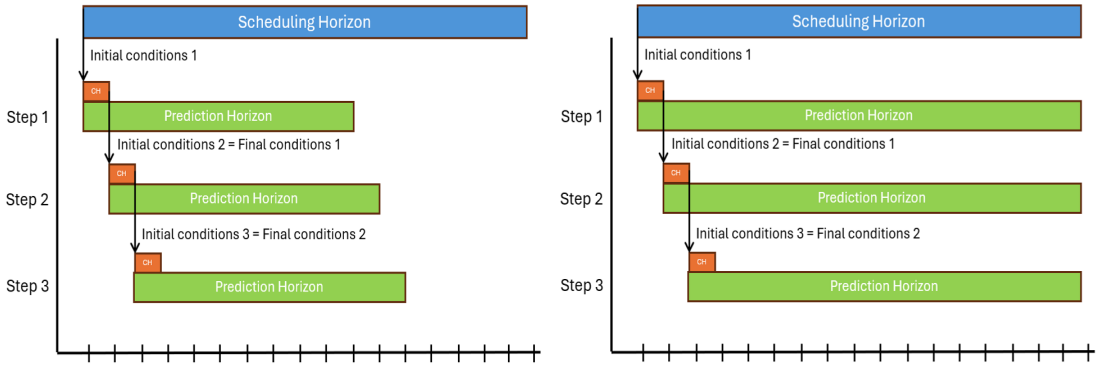


Figure 4.2.2: SDDP Scenario tree for three stages

The solution approach is based on the approximation of expected future cost functions by piecewise linear functions, instead of using an interpolated table done in SDP [68]. These approximate functions are obtained from the dual solutions of the optimization problem at each stage and can be interpreted as Benders cuts in a stochastic, multistage decomposition algorithm [65]. SDDP is a recursive algorithm, which employs a forward and backward pass. The forward pass moves down the scenario three picking scenarios at random. The information about the state variables is carried over from stage to stage. In the backward pass, all of the possible scenarios are run and Benders cuts are created, representing the future cost of the state variable. A drawback with SDDP is that non-convexities can not be easily treated in the modeling [70]. The non-convexities can in some cases be handled by either making approximations or combining SDDP with SDP.

4.3 Model Predictive Control Optimization

Model predictive control (MPC), also known as receding horizon control, moving horizon control, or rolling horizon control, is a feedback control technique that solves an optimization problem at each time step to determine the optimal control actions over a fixed time horizon [71]. Normally MPC consists of three stages; a model to predict future outputs, calculate control actions to optimize performance, and apply a horizon strategy to dynamically update new information in the optimization process when new information becomes available [72]. MPC is a well used feedback control technique and has been used since 1980 [71]. MPC algorithms often solve problems under uncertainty or large-scale optimization problems [73], and are often used to solve MSSP problems [74]. MPC algorithms have among other things become important in the operation of energy systems because the optimal operation must account for the uncertainty of consumer's behavior, weather conditions, and power generation from renewable energy sources [75]. The reason why the feedback control technique goes by so many names is due to how the update of new information is done. The control techniques are commonly divided into receding horizon (RH) and rolling horizon (ROH) [76], presented in Figure 4.3.1



(a) Illustration of a rolling horizon framework **(b)** Illustration of a receding horizon framework

Figure 4.3.1: MPC control techniques, inspired from [73]

Figure 4.3.1a presents a ROH approach, consisting of three different horizons: the scheduling horizon (SH), the prediction horizon (PH), and the control horizon (CH) [73]. SH represents the total operational period and is decomposed into equal time intervals. The length of the PH is defined by the accuracy of the prediction data for each subsequent time step, and the data can be considered to have some certainty [71]. The CH has the same initial conditions as the PH but covers a shorter duration, often with a time step equal to one time interval. The solved decision variables in the CH then used as input in the PH, and are presented in the final solved model.

Figure 4.3.1a showcases the dynamic updating of horizons for a RH-framework. In this approach, the PH is equal to the SH. A long PH will allow more flexibility in optimizing future control actions but can result in more uncertainty in the data. A short PH will on the other hand lower the CHs ability to plan ahead effectively.

For both of the control techniques, when the optimization problem is solved for each time step in the CH, the initial values are decided at the first time step, and the algorithm optimizes the optimization problem with a forecast for the next steps. From the solved model, the optimized decision variables will be given as initial values for the next time step. The algorithm will then solve the new optimization problem with the new time horizon.

Optimization Literature Review

4.4 Model approaches

In this section, studies on EV charging modeling approaches and methodologies are presented and compared. The goal is to find and compare the different factors, assumptions, and strategies that are used when modeling EV charging. It is also investigated how previous studies have implemented methods to optimize the short-term real-time pricing and and the long-term impact of the demand charge.

4.4.1 Model approaches for optimizing EV charging

One of the challenges in modeling EVs is that each EV represents unique data because each EV is dependent on the behavior of the EV owner. When modeling an EVPL, the ideal is to have specific individual data on each EV for all time steps. A comparison is made on the distinctions between an aggregated model and a non-aggregated model.

In the study done by J. Zhoa [56], an aggregated model for EV driving patterns is used to simplify the optimization model. The decision to use an aggregated model is driven by practical considerations, computational efficiency, contractual arrangements with EV owners, and the need for effective representation and scheduling strategies in the electricity market, especially when dealing with uncertainty and risk. In Zhoas study, the following is stated:

The computation burden can be very heavy if each individual EV is dispatched directly by the system operators. Therefore an aggregated model is developed in the study to condense individual driving pattern data into a simplified model, in analogy to a virtual battery with the total capacity, charging, and discharging ability changing over time. In practice, the EV aggregator makes a long-term contract with EV owners. [56].

It was found that when V2G is included, an aggregated model was often used for the EVPL [77] [56]. However, in the study done by Y. Chai et al. [41], V2G was included, and a non-aggregated model is used. To determine the charging and discharging schedule of the EV, they predict day-ahead data of the EVs. The required data was arrival time, departure time, and the desired SOC. EV behavior is defined as vehicle occupancy, the arrival- and departure time. The EV owners might not be able to stay idle until the announced departure time, which is a concern for the building owners who require the full commitment of the EV owners to participate in the V2G event. In the study, EV owners who practice flexible working hours are unable to leave the building earlier than the expected time slot at the desired SOC value. It is stated that the coordination between building and EV owners is challenging due to the stochastic nature of load and EV mobility, and therefore they require a full commitment of the EV owners.

Through the previous studies it is found that using a non-aggregated model requires EV input data to be provided in advance, or many assumptions have to be made. When planning optimization for the EVPL, full commitment from the EV owners is necessary. Additionally, models implementing V2G often use an aggregated approach to avoid the requirement of detailed information on each EV.

4.4.2 Model predictive control and control techniques

The goal of this section is to review the feedback control methods used in other literature and assess the possibilities of different control techniques. Other methods used for real-time and feedback control are Minimum Variance, Smith Prediction, Generalized Minimum Variance, and Pole Placement. With the right parameters, these methods can perform very well,

however, in journal [72], several weaknesses are pointed out. These methods face challenges such as increased computational effort and sensitivity to errors, difficulty handling some types of systems, high control effort, and lack of robustness against variable or unknown delays. Modifications of Generalized Minimum Variance led to the development of MPC, and journal [72] states that MPC is preferred because it offers better performance and control.

In Section 4.3, RH and ROH were presented as two control techniques for MPC. Article [76] points out that when using RH optimization, the system can update the available information and obtain a better response. A RH-framework was implemented in a study done by I. Skoglund et al. [78] to schedule the charging and discharging of batteries in commercial buildings while also considering the stochastic nature of the data and the monthly DCC. Using the RH-framework to operate the battery, the total system cost was reduced up to 17.8%. In [79], an RH approach was used on a deterministic multi objective optimization problem to find the optimal weighting between diesel fuel cost and battery wear cost. The RH approach resulted in a long run time, as an optimization procedure must be done for each time step. However, they also found that the results obtained from RH were more effective and feasible when compared with other conventional optimization methods, as RH offers a significant advantage by accounting for the uncertainties and intermittent nature of renewable energy resources.

In a study done by M. Tvedt [80], a ROH-framework was used to solve a linear optimization problem to minimize the total cost of operation, using flexible assets. The study did not consider the monthly demand charge which meant that using ROH was therefore suitable. The study points out that grid tariff could be interesting to include in further work, however, this would require changes in how the data is updated in the PH. Article [81] uses a ROH-framework with an 8 hour PH to optimize EV charging. The study suggests implementing dynamic horizons to optimize EV charging better.

In the article written by K. Baker et al., a stochastic RH approach is used to plan for energy storage systems from an operational point of view [82]. By integrating a stochastic RH approach, uncertainty can be added to the optimization. When using an MPC approach on a stochastic optimization problem, the RH algorithm had to consider all time horizons and scenarios simultaneously. This made the problem very complex and computationally heavy. In their future work, they suggested that integrating scenario-based decomposition methods, like Benders' decomposition, could expand the scope to include more scenarios in the problem.

The studies show that a feedback control method can lead to cost savings and improved system reliability, making it an important tool for analyzing and optimizing real-time energy management strategies. Study [82] also states that a decomposition method can be of value in combination with RH. A ROH approach is seen to struggle to consider the monthly demand charge. Dynamic horizons is suggested to provide better real-time optimization of EV charging.

4.4.3 Stochastic dual dynamic programming

SDDP is commonly used in hydropower and hydropower modeling as it avoids the necessity of discretizing the state values such as inflow and reservoir volume, making it possible to model several reservoirs at once. The issue with non-convexity mentioned in 4.2.2 within hydro generation modeling is explained and illustrated in [83]. The non-convexity is avoided in [84] by modeling the before non-convex price state as a Markov chain using discrete spaces similar to what is done in SDP.

SDDP is also used as a decomposition technique for applications outside of hydrothermal models. In [85], an SDDP algorithm is used to model an integrated photovoltaic and energy storage hybrid system in a household. The MSSP presented had 10 scenarios for each of the 96 stages, which would be computationally unfeasible if discretized. Therefore, SDDP is used to avoid the well-known curse of dimensionality associated with stochastic problems. In [86] an SDDP approach is developed to model distributed energy resources as an aggregator. The reason for choosing SDDP instead of other decomposition methods is SDDPs ability to refine solution quality in the areas where the solution is most likely to occur, together with SDDPs ability to handle multiple state variables with lower computational effort.

In [87] SDDP is used mainly to consider both short-term costs and the price signals of long-term costs of the energy use in a building. The future cost function created by SDDP managed to represent the long-term future cost implication of the raising the peak, although not with ideal accuracy.

4.4.4 Combining receding horizon and stochastic dual dynamic programming

As suggested by K. Baker et al. in Section 4.4.2 [82], there are incentives to integrate scenario based decomposition methods with the RH framework. Using the long term planning from the expected future value functions of the decomposition techniques could provide value to the short-term optimization in RH.

An article written by A. Papavasiliou et al. [88] proposes an MSSP model for transmission-constrained economic dispatch, emphasizing real-time optimization of storage deployment with an uncertain multiarea renewable production. The authors state that much of the existing literature analyzes storage values through two-stage day-ahead models or economic dispatch models with limited look-ahead, which they find too optimistic. A gain of 1.1% is found when using the stochastic real-time approach rather than a deterministic forecast.

SDDP together with an RH-framework was incorporated in a study done by F. Hafiz et al. [89] to operate a battery system. The SDDP algorithm was used to solve an MSSP, that iteratively constructed a segmented linear approximation of the future costs function. The RH was used to update the forecast of the load and solar generation, which improves the optimal decision-making process for the battery system. SDDP provided set points for the battery

SOC, and the RH-framework operated the battery in real-time to reduce the cost of operation. Here, significant peak hour savings are found compared to a heuristic method.

The integration of RH control with SDDP entails a promising approach for optimizing energy systems under uncertainty. By utilizing the long-term planning capabilities of SDDP with uncertainty and the real-time strategy of RH, improvements in decision making and operational efficiency can be achieved.

Chapter 5

Method

5.1 Model Description

This chapter introduces the model that was created to minimize the electricity costs for an office building with an EVPL. The model setup and methodology are presented in detail, covering the motivations and assumptions for the choices made.

5.1.1 Model overview

The modeling of the operation of the office building with a connected EVPL requires detailed information on a short-term basis to make hourly operational decisions. The system is also modeled over one month to consider the monthly grid tariff costs. The system is investigated for participation in an IBDRP and is also subject to day-ahead spot prices, which require dynamic updates of information. A version of a RH-framework is introduced which introduces dynamic horizons that dynamically update the information available in the optimization problem. The information available to the EVPL is on a daily basis, making it beneficial to decompose the monthly problem into daily stages. The monthly peak is the only state variable of the model, and information about the monthly peak is carried over between the daily stages.

As the demand charge is based on the peak import hour in a month, the long-term impact of the peak level in the model needs to be considered for an optimal result. The RH-algorithm on its own would struggle to optimize for the final peak value of the month, as detailed information is only provided on a daily stage-wise basis. SDDP is therefore introduced to create future cost curves containing cost information about the monthly peak. The SDDP-framework uses the same decomposed monthly problem, but also introduces uncertainty in the modeling of the office-building load for a more realistic approach.

In Figure 5.1.1, an overview of the whole model is presented. The *Input Data* block organizes

and puts into action the data used in the model. The *Model Setup* block includes the general optimization problem for the SDDP- and RH-framework. The SDDP-framework produces marginal cost curves in each iteration, which are then passed to *Model Setup*. With the information from the *Model Setup* including the future cost curves, the RH-framework can generate real-time results for the output.

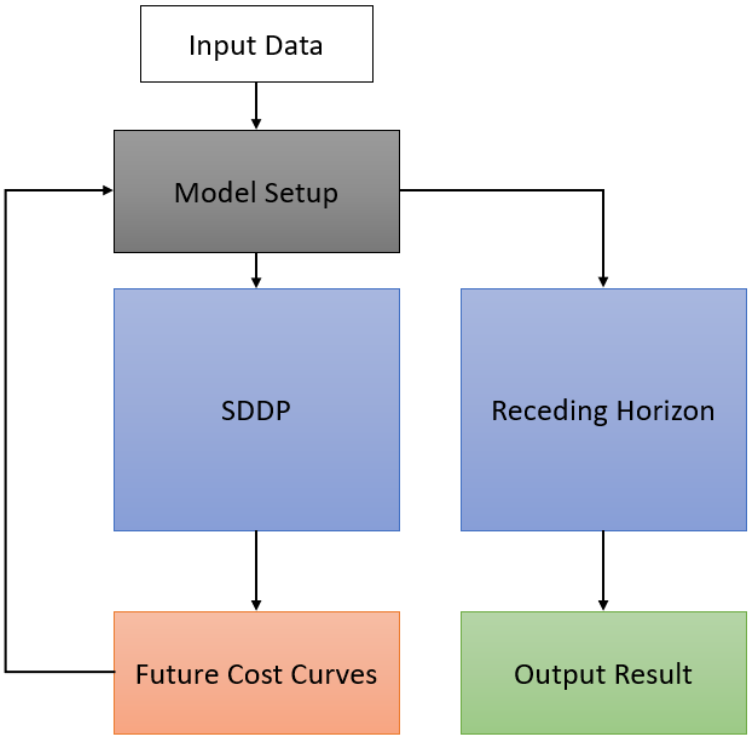


Figure 5.1.1: Illustration of the method, inspired from [90]

5.1.2 Model setup

Figure 5.1.2 showcases the structure of the system that is modeled, and the subsequent decomposed optimization problem which is used in model-framework. Within the measurement point, there is an office building, an EVPL, and a V2G-enabled charging system. The system is assumed to have an unlimited capacity to import and export power between the measurement point and the electricity grid, represented by p_t^{imp} and p_t^{exp} respectively. Between the measurement point and EVPL, p_t^{EV}, $p_t^{V2G,ch}$ and $p_t^{V2G,dch}$ are the variables which represent the power flow. Lastly, P_t^O represents the power sent to the office building.

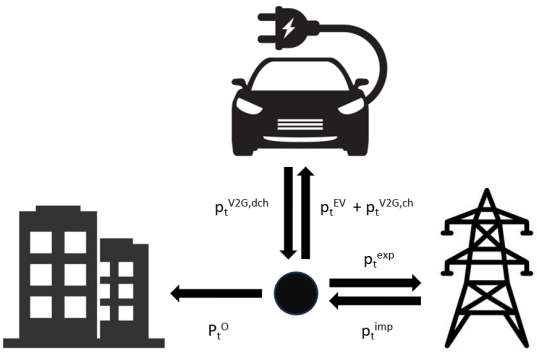


Figure 5.1.2: Illustration of the system

The model assumes aggregated EVPL input data, as individual modeling of the EVs requires large amounts of data on the individual behavior of each EV. Figure 5.1.3 showcases how each EVs decision variables are modeled as an aggregated "black box" and combined to represent one EV. The aggregated EV has the same properties as each individual EV, with the ability to discharge and charge.

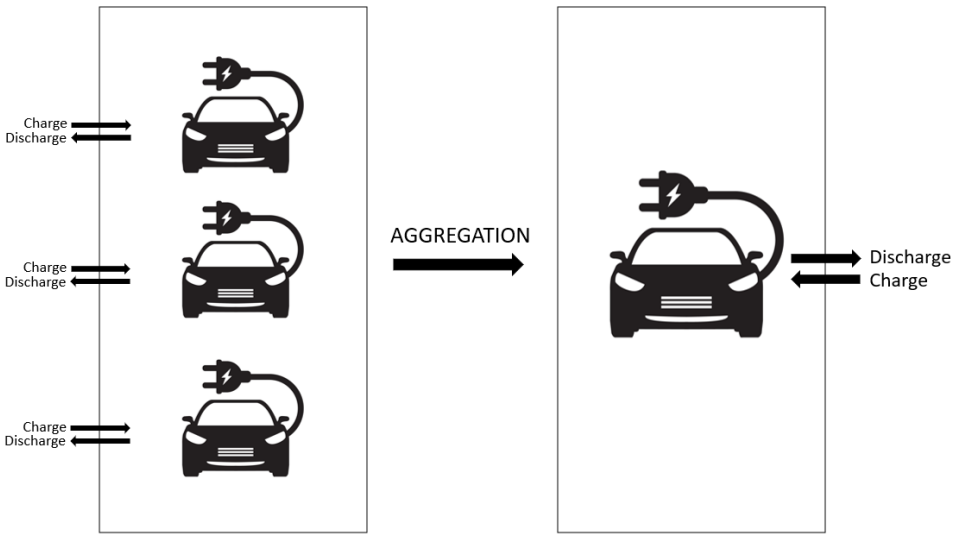


Figure 5.1.3: EV Aggregation

5.1.3 Notation

The whole notation for the thesis is presented here, including sets, parameters, and variables.

Sets

T : Set of hours in time period, indexed by t

D : Set of days in the time period, indexed by d

C : Set of cuts, indexed by c

G : Set of decision stages, indexed by g

S : Set of scenarios, indexed by s

Parameters

P_d^{EV} : EV demand at the EVPL at day d $[kWh]$

$P_d^{EV,init}$: Initial energy imported at the EVPL at day d $[kWh]$

P_t^{O} : Office building energy demand at time t $[kWh/h]$

$P_t^{EV,cap}$: EVPL charging capacity at time t $[kWh/h]$

C_t : Cost of power at time t $[NOK/kWh]$

C^{vol} : Volumetric Cost of energy $[NOK/kWh]$

C^{peak} : Monthly demand charge cost $[NOK/kWh/h]$

$C^{not\,delivered}$: Cost of not delivering energy to EVPL $[NOK/kWh]$

$\overline{p_c^{peak}}$: Peak power for cut c $[kWh/h]$

ϕ_c : Objective value for cut c $[NOK]$

λ_c : Dual value for cut c $[NOK/kWh/h]$

$P^{peak,init}$: Initial peak-import power $[kWh/h]$

$P^{EV,Thr}$: EV charging limit by IBDR-signal $[kWh/h]$

$P^{V2G,Thr}$: V2G charging limit by IBDR-signal $[kWh/h]$

ENS_d^{init} : Initial energy not supplied to the EVPL at day d $[kWh]$

$P_d^{V2G,ch,init}$: Initial charged V2G energy at the EVPL at day d $[kWh]$

$P_d^{V2G,dch,init}$: Initial discharged V2G energy at the EVPL at day d $[kWh]$

η : V2G charging and discharging effency $[\%]$

Variables

p_t^{imp} : Energy imported at time t $[kWh/h]$

p_t^{EV} : EV charging quantity at time t $[kWh/h]$

p^{peak} : Peak of imported energy $[kWh/h]$

α : Future cost of operation from linear cuts $[NOK]$

$p_t^{V2G,dch}$: Discharged energy from EVPL using V2G at time t $[kWh/h]$

$p_t^{V2G,ch}$: Charged energy from EVPL for V2G at time t $[kWh/h]$

p_t^{exp} : Exported energy to the grid at time t $[kWh/h]$

ens_t : Energy not supplied to the EVPL at time t $[kWh]$

5.1.4 Decomposed optimization problem

5.1.4.1 Objective function

The objective in the optimization problem is to minimize the costs associated with the electricity use of the office building and EVPL. The objective function is shown in equations (5.1) - (5.4). The costs associated with electricity use are the market price for electricity and the volumetric costs from the grid tariff (equation 5.1). The electricity sold back to the grid through the use of V2G (equation 5.2) is considered profit and therefore has a negative sign. The cost of not delivering the requested charging (equation 5.3). The fixed grid-tariff costs and the future costs (equation 5.4).

$$OBJ = \sum_{t \in \mathcal{T}} (C_t + C^{vol}) \cdot p_t^{imp} \tag{5.1}$$

$$- \sum_{t \in \mathcal{T}} C_t \cdot p_t^{exp} \tag{5.2}$$

$$+ \sum_{t \in \mathcal{T}} ens_t \cdot C^{not\,delivered} \tag{5.3}$$

$$+ C_{fixed} + \alpha \tag{5.4}$$

5.1.4.2 Constraints

The model has implemented several constraints which are needed for the model to operate as realistically as possible. The constraints presented are divided into sections containing the different parts of the system.

Loads
The hourly load balance is given in Equation (5.5), where the import of electricity from the grid must cover hourly office building demand and the hourly energy used for EV charging.

$$p_t^{imp} = P_t^O + p_t^{EV} \quad \forall t \tag{5.5}$$

The input data used for the office building demand and the EV charging is based on real-life measurements with the losses already included. The efficiency factor η is therefore not included to model the losses associated with the imported energy.

$$P_d^{EV} = \sum_{t \in T_d} (p_t^{EV} + ens_t) + P_d^{EV,init} + ENS_d^{init} \quad \forall d \in D \tag{5.6}$$

The daily aggregated EV-charging demand represented by the parameter P_d^{EV} is presented in Equation (5.6). This constraint is included to ensure the daily EV charging demand is met. $P_d^{EV,init}$ and ENS_d^{init} are used by the RH algorithm to represent the charging and the ENS which is previously done on day d, respectively.

Electric vehicle parking lot

The power flow into the EVPL is restricted by the main fuse, which limits how much power can be drawn by the parking lot. This parameter, $P_t^{EV,cap}$, also plays a role in regulating the hours at which the charging station is available for charging. Equation (5.7) denotes the time-dependent charging capacity of the EVPL.

$$0 \leq p_t^{EV} \leq P_t^{EV,cap} \quad \forall t \tag{5.7}$$

Vehicle to grid

The V2G technology lets the system discharge the EVPL to either cover the office building load or export to the grid. As the V2G charging and discharging are not based on a parameter from real-life measurements, losses are included. The efficiency constant for discharging and charging are equal and given as η.

With the EVPL being modeled as an aggregated model, the state of charge of the EVs is not considered, and it is assumed that the EVs always have enough energy as a group to be able to discharge if needed. Equation (5.8) ensures that the power flow does not exceed the hourly power that can be drawn from or to the EVPL by using the parameter $P_t^{EV,cap}$.

$$0 \leq p_t^{V2G,dch} + p_t^{V2G,ch} + p_t^{EV} \leq P_t^{EV,\text{cap}} \quad \forall t \tag{5.8}$$

The load balance for V2G is shown in Equation (5.9). $p_t^{V2G,dch}$ represents the discharging from the EVPL, and p_t^{exp} represents the energy which is exported to the grid. There is no restriction put in place that forbids the model to charge and discharge at the same time, as introducing a binary variable would significantly complicate the algorithm. This is of no concern because of the losses included in the charging and discharging when using V2G.

$$p_t^{imp} - p_t^{exp} = P_t^O + p_t^{EV} - p_t^{V2G,dch} \cdot \eta + p_t^{V2G,ch} \quad \forall t \tag{5.9}$$

The constraint shown in Equation (5.10) ensures that the energy that is discharged from the EVPL through V2G is recouped and charged back up by the end of the day.

$$\sum_{t \in T_d} p_t^{V2G,ch} + P_d^{V2G,ch,init} = (\sum_{t \in T_d} p_t^{V2G,dch} + P_d^{V2G,dch,init}) \cdot \frac{1}{\eta} \quad \forall d \in D \tag{5.10}$$

Grid tariff

The Grid Tariff which is explained in Section 2.2.3 has a component which is based on the peak power usage behind the meter on a monthly basis. The state variable of the system, p^{peak}, needs to carried over in between days. Equation (5.11) ensures that the p^{peak} used in both SDDP and RH is the highest value achieved, by comparing the peak value against the imported energy each hour.

$$p^{\text{peak}} \geq p_t^{\text{imp}} \quad \forall t \tag{5.11}$$

If the value of the imported energy p_t^{imp} is higher than p^{peak}, then p^{peak} is updated with the new peak value. Further, Equation (5.12) ensures that the state variable p^{peak} is accounted for and carried over from day to day.

$$p^{\text{peak}} \geq P^{peak,init} \tag{5.12}$$

The future cost of operation α, influenced by the highest peak import, is represented by linear cuts from the SDDP-framework in Equation (5.13). The cut generation is explained further in Section 5.2.1.

$$\alpha_c \geq \Phi_c + \lambda_c \times (p^{peak} - \overline{p_c^{peak}}) \quad \forall c \tag{5.13}$$

Throttling
To model the throttling signals given by the IBDRP which is revealed during the RH-algorithm, a constraint is introduced in Equation (5.14) which limits the power available to use in the EVPL for a given hour s.

$$p_s^{EV} \leq P^{EV,Thr} \tag{5.14}$$

Equation (5.15) represents the limit for the minimum discharge needed from the V2G system when the throttling value exceeds $P_t^{EV,cap}$. This constraint makes it possible for the system to deliver power back to the grid in throttling hours, which doubles the throttling capacity. $P^{EV,Thr}$ is a parameter used as input to the model when V2G is included in the system, which can have both negative and non-negative values. As long as $P^{EV,Thr}$ is positive, $p_s^{V2G,dch}$ is zero.

$$p_s^{EV} - p_s^{V2G,dch} \leq P^{EV,Thr} \tag{5.15}$$

5.2 Stochastic Dual Dynamic Programming Framework

The strategy phase section can be interpreted as the left part of Figure 5.1.1, encompassing the SDDP algorithm and the creation of the future cost curves. This section employs the model explained in Section 5.1 to create future cost curves. These future cost curves are used by the real-time simulation framework explained in Section 5.3.

5.2.1 Stochastic dual dynamic programming algorithm

A brief theoretical explanation of SDDP explaining the structure and upsides/downsides was performed in Section 4.2.2. Here, a more detailed explanation of the algorithm used in the thesis is presented. The SDDP algorithm is shown in Algorithm 1, and is based on the algorithm presented in **Paper I** but modified to fit this thesis. The algorithm makes use of the optimization model presented in Section 5.1.2 to create linear future cost cuts. The uncertainty of the stochastic framework is implemented in the daily office building load

demand. The goal of the SDDP algorithm is to contribute in the operational planning with the ability to consider the future costs of the monthly peak, which can help the real-time operational planning with a better short-term and long-term planning. The SDDP algorithm does not factor the IBDR-signals presented in Section 5.1.4.2, as it is assumed that the long-term planning does not plan for IBDR-signals to occur.

Algorithm 1: The SDDP algorithm

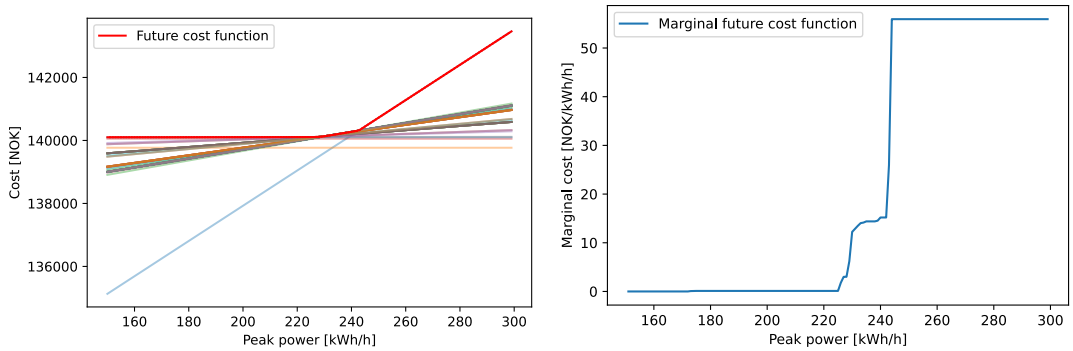
1 **while** *run = True* **do**
2 $P_0^{\text{peak,init}} = 0$, $i+ = 1$;
3 **for** $n = 0, 1, 2, \ldots, G-1, G$ **do**
4 Sample a scenario s_n;
5 $P_n^{\text{peak,init}} = p_{n-1}^{\text{Peak}}, P_t^O \leftarrow P_t^{O,s}$for t ∈ T;
6 $p_n^{peak} \leftarrow Optimize(1-7)$;
7 **for** $n = G, G-1, \ldots, 1, 0$ **do**
8 $P_n^{\text{peak,init}} = p_{n-1}^{\text{Peak}}$;
9 **for** $s \in S$ **do**
10 $P_t^O \leftarrow P_{t,s}^O$for t ∈ T;
11 $\phi_n^s, \lambda_n^s, \leftarrow Optimize(1-7)$;
12 $\phi_{n,i} = \sum_{s \in \text{S}} \rho_{d,s} \cdot \phi_n^s, \lambda_{n,i} = \sum_{s \in \text{S}} \rho_{d,s} \cdot \lambda_n^s, \overline{p_{n,i}^{peak}} = \sum_{s \in \text{S}} \rho_{d,s} \cdot P_n^{\text{peak,init}}$
13 **if** *new cuts not unique* **then**
14 Convergence achieved, $run = False$;

The whole algorithm is inside of a while loop, shown in line 1, which runs until the convergence criterion at line 13 and 14 is met. The convergence criterion for the algorithm is based on the cuts which are generated. The algorithm checks if all of the cuts generated in the last iteration are equal within a certain margin of $\pm\ \epsilon = 0.01$ to the cuts created in any of the previous iterations. All of the cuts for all of the stages and all of the scenarios have to be equal for the algorithm to recognize them as equal. For the algorithm to stop the while loop, the algorithm has to output already existing cuts three times.

The algorithm is composed of a forward loop (lines 3-6) and a backward loop (lines 7-12), as mentioned in Section 4.2.2. Before the simulation, on line 2, $P_0^{\text{peak,init}}$ is initialized to a value of 0. The forward loops computes the decomposed optimization problem for all stages, drawing a random scenario for the office load $P_{t,s}^O$ at each stage, starting at the first stage $n = 0$. After each stage, the state variable p_{n-1}^{Peak} is saved and used as an input parameter $P_n^{\text{peak,init}}$ in the next stage. In the backward loop, starting at the last stage $n = G$, the decomposed optimization problem is computed for all possible scenarios. Each decomposed problem outputs a set of parameters, most notably the objective function value ϕ_n^s, the dual value λ_n^s of, and the initial peak $P_n^{\text{peak,init}}$. The parameters are then weighted by their respective scenario to generate a weighted linear cut for the given stage in a single-cut approach, shown in line 12, using Equation 5.12 to create the cuts. The weighting of the cuts is based on the respective probabilities for the stage transition, providing one weighted cut for each scenario the model is currently in.

5.2.2 Future cost curves

The resulting cuts from Algorithm 1 represent the future cost of the peak value for the given stage and scenario. In Figure 5.2.1, two representations of the cuts are shown. In (a), the linear cuts for an arbitrary day are plotted for example purposes. These linear cuts are used as input parameters to the RH-framework to more accurately weigh the future cost of increasing the peak value. The linear cuts that restrict the state space can then be represented as a linear piecewise function, shown as the red line in plot (a). To easier visualize the cuts and their effect on the peak values later in the thesis, the marginal is taken of the piecewise linear function, presented in (b).



(a) Linear cuts with piecewise linear future cost function

(b) Marginal future cost function

Figure 5.2.1: Linear cuts and marginal cost curves

5.3 Receding Horizon Algorithm and Framework

As mentioned previously, a RH approach with dynamic horizons is implemented in this study to be able to account for the sudden IBDR activaion signals and also optimize the operation of the system on a hourly basis. The framework for the approach is presented in this section, explaining how the information is revealed to the optimization problem in the various horizons.

The load data for the office building is assumed to be known for the entire SH, and the spot price is revealed only once daily. While a ROH approach could work for this, it would not be able to account for the monthly demand charge. Alternatively, a RH approach with a PH equal to one month could consider the monthly demand charge, but a constant PH can not include information about the future daily spot prices. Therefore, this methodology presents a combination of both approaches, referred to as RH, which uses future cost information from SDDP to handle the long-term implications of the demand charge.

When implementing the RH algorithm, the first stage is to define the initial state of the system. First, the variables in the model are initialized, including the initial variable P^{init}. The RH-framework considers three different horizons, where the SH considers the overall operational period. The CH represents the current hour of operation that is to be determined

and moves forward for each step. Lastly, the PH represents the period within the optimization problem containing detailed information about the data related to spot prices, office demand, and EV demand.

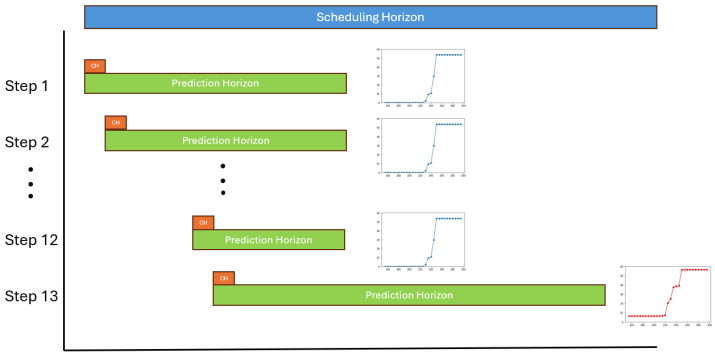


Figure 5.3.1: Receding horizon framework. Figure inspired from **Paper I**.

Figure 5.3.1 shows how the linear cuts derived from the SDDP-framework describe the information beyond the PH. The linear cuts give the PH a simplified representation of the rest of the SH, including the monthly demand charge. The methodology for this thesis presents a new PH to the RH-framework at each time step. When the spot prices for the next day are revealed, the PH is expanded, revealing information about the next day of operation. At the same time, the linear cuts from the SDDP-framework are replaced so the PH can consider the cuts from the next day of operation.

Within the RH framework, operational decisions are made at each time-step t in the CH, while the PH is continuously adjusted based on the operating day, as illustrated in Figure 5.3.1. At each step, the highest achieved peak p^{peak} so far in the optimization is stored in $P^{peak,init}$ together with the accumulated EV charging demand for the current day $P_d^{EV,init}$ and the accumulated ENS for the current day ENS_d^{init}. The stored values are then used as input for the next time-step *t+1* within the RH framework. This process is repeated until the RH-framework reaches the final time step of the SH. When integrating V2G into the model, the accumulated V2G charging $P_d^{V2G,ch,init}$ and discharging $P_d^{V2G,dch,init}$ are also stored for the current day, and provided as inputs for the next time-step. Figure 5.3.2 presents a block diagram of the RH algorithm used in this thesis.

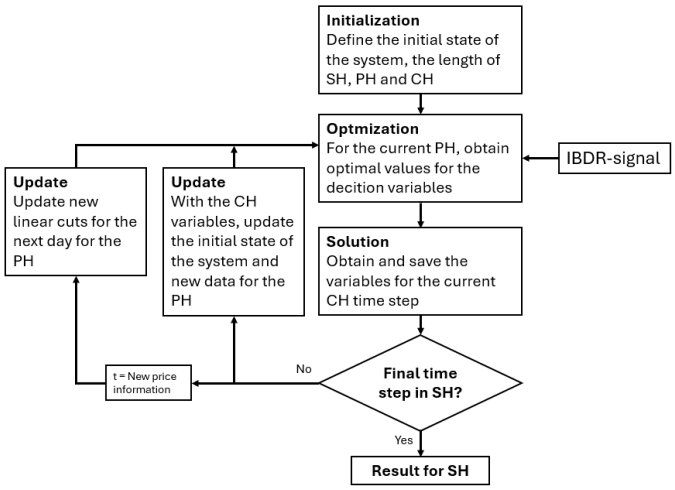


Figure 5.3.2: Flow chart showing a overview of the RH information update-process

The IBDR-signals are included in the RH-framework by including Equation (5.14) in the optimization problem when a notice is given. The notice time is an input parameter to the RH-framework. The notice time before activation determines at which time-step Equation (5.14) is revealed to the problem. The constraint limits the EV charging capacity for activation hour s, and the RH-framework must consider this limitation until hour s has passed. When hour s has passed, the constraint is removed until a new IBDR-signal is given. This also yields Equation (5.14) when V2G is integrated, where there is an IBDR-signal that requires a throttling bigger than what the p_t^{EV} can provide.

Chapter 6

Case Study

The primary goal of this study is to investigate how the operation of a flexible EVPL can reduce cost of operation and peak consumption for an office building while offering grid services. To do this, a case study is developed to investigate how the different technologies and IBDRP signals influence the decisions and the resulting strategies. This chapter gives an introduction to the cases and scenarios, together with an analysis of the input data and the parameters used in the optimizing frameworks.

6.1 Python

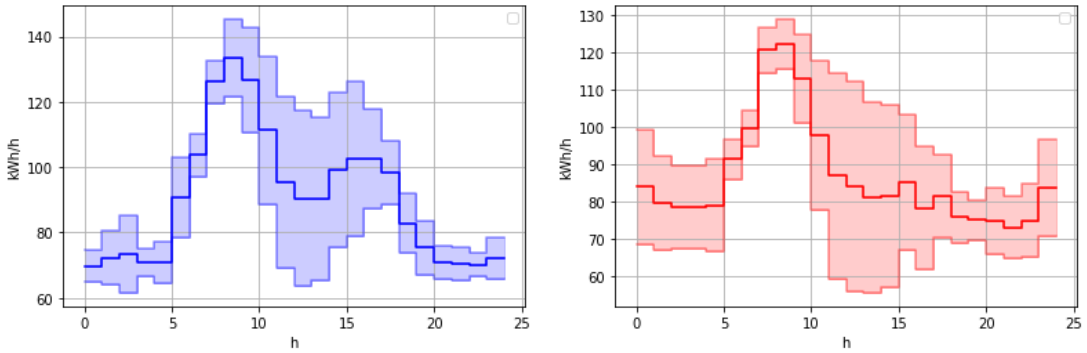
Python is used as the programming language when solving the optimization problems. Python is an open-source programming language with support for a Python standard library and community-contributed modules [91]. Pyomo, a Python-based, open-source optimization modeling software package was used to develop the optimization model [92].

6.2 Input Data

This section includes the relevant input data used in the thesis. Lede AS and the partners in the Strømfleks project [1] have provided load data for the office building and the EVPL, located in the southeastern part of Norway (NO2 price zone), and the measurement data used in this thesis is from 2022. This data was divided into main-meter readings which encompassed all of the load data, and EVPL data. What is known as office building data in this case is the main meter readings without the EVPL data. A few non-values were interpolated to create a continuous dataset, and the accumulated dataset was restructured to a sequential dataset. The spot price data used is historical day-ahead data for the respective months. A short analysis is performed on the input data to help create relevant cases.

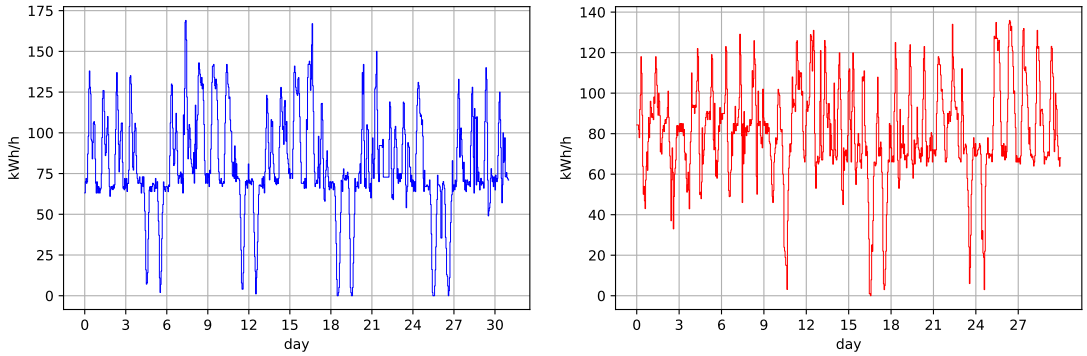
6.2.1 Office building demand

Historical hourly load data from the office building is used as an input parameter and is considered inflexible. Shown in Figure 6.2.1 is the average hourly office building load for weekdays. Weekends are filtered out of the plot to show only the office building load which is relevant for peak optimizing and EV charging. The hourly office-building load over a month is presented in Figure 6.2.2 for two representative months, namely March and September.



(a) March 2022 **(b)** September 2022

Figure 6.2.1: Average hourly office-building load with standard deviation for weekdays



(a) March 2022 **(b)** September 2022

Figure 6.2.2: Hourly office-building load for a month

6.2.2 Electric vehicle parking lot

In reality the EV charging demand for the EVPL varies between each workday, but for this thesis, the historical data for weekdays is averaged so that the EVPL load demand is equal for each day throughout the month. The average EVPL load, denoted as P_d^{EV} in the model, is presented in Table 6.2.1.

The EVPL has a dynamic charging limit, based on EV arrival times, and a 125 kW fuse which limits the transfer capacity to the EVPL. The ramping of the EVPL charging capacity is based on the arrival times from [93]. The EVPL is assumed to have 50 chargers, all with an average

Table 6.2.1: Average EVPL load with standard deviation in 2022

Month	Energy (kWh)
March	670.7 ± 53.7
September	694.7 ± 96.8

charging power of 5 kW. It is assumed that no charging occurs in the weekend, which is primarily the case from the given input-data. The resulting charging capacity is shown in Table 6.2.2.

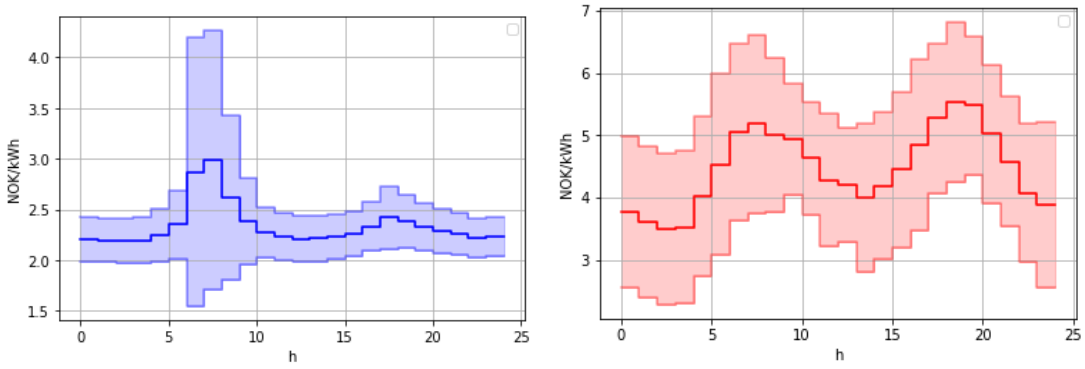
Table 6.2.2: Hourly power capacity for EVPL

Hour of the day	# of EVs	EV_{cap} (kW)
0-5	0	0
6	5	25
7	15	75
8-14	50	125
15	20	100
16	10	50
17-23	0	0

Implementing V2G to the system, the EV capacity in Table 6.2.2 is also implemented on the V2G discharging. The V2G charging and discharging is not based on historical data and therefore losses are added in the model. The losses are implemented on the import/export so that the maximum transfer capacity to the EVPL is still at the same level as charging without V2G. It is assumed that the charging and discharging has the same efficiency of $\eta = 86.5\%$, which is based on data from [94]. As there is no data on the individual state of charge of the arriving EVs, it is assumed that there is always energy available in the EVPL to be discharged. When exporting the discharged energy to the grid, the volumetric cost from the grid tariff is not included, which incentivizes the system to use the discharged energy locally instead of exporting it to the grid. It is assumed that all the EVs present at the EVPL are capable of bi-directional charging.

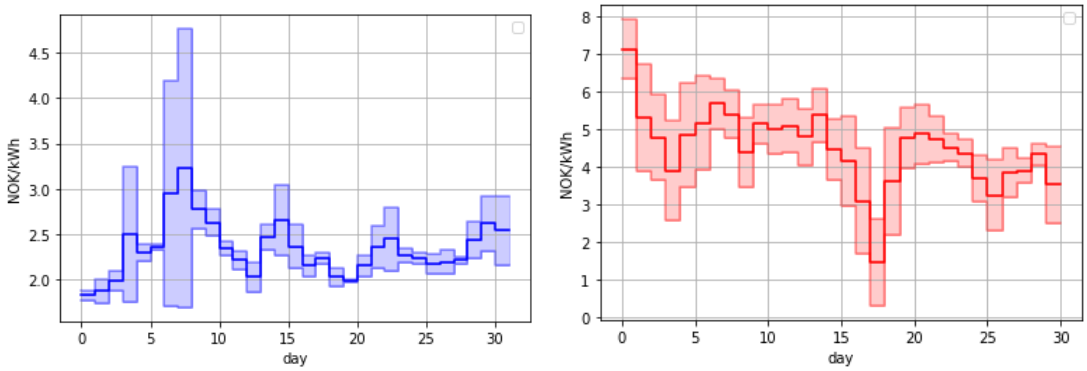
6.2.3 Electricity price

The price of electricity is based on Nord Pools day-ahead prices [14], which are a result of the Nord Pools day-ahead implicit auction market. As the office building is located in the southeastern part of Norway, the data is gathered from the price region of NO2 in Norway. The average hourly spot price profile is plotted in Figure 6.2.3, and the average daily spot price is plotted in Figure 6.2.4. 2022 was quite an abnormal energy year and therefore the spot prices might differ from what is expected, with high volatility and high average prices.



(a) March 2022 **(b)** September 2022

Figure 6.2.3: Average hourly spot price with standard deviation



(a) March 2022 **(b)** September 2022

Figure 6.2.4: Average daily spot price with standard deviation

6.2.4 Grid tariff

The grid tariff used by the office building in this model is by the DSO Lede, in our case for large customers [95], presented in Table 6.2.3.

The high-voltage tariff is chosen for the system. Given that there is no information available on the reactive power consumption of the system, the reactive power cost is neglected. A monthly DCC for winter months at 57.16 NOK/kW and 41,9 NOK/kW for summer months is used on the highest achieved single-hour load. The volumetric costs used are presented below.

$$C^{vol}_{jan-mars} = 0.0475 + 0.1189 = 0.1664 \quad [NOK/kWh]$$
$$C^{vol}_{april-sep} = 0.03 + 0.2055 = 0.2355 \quad [NOK/kWh]$$

The fixed monthly cost is:

$$C^{Fixed} = 625 + (\frac{800}{12}) = 691.67 \quad [NOK/month]$$

Table 6.2.3: Lede grid tariff cost, VAT included [95]

Categories	**High voltage**
Fixed cost	625 NOK/mnd
Energy charge winter (oct-march)	4,75 øre/kWh
Energy charge summer (april-sept)	3 øre/kWh
Active power winter (oct-march)	57,16 NOK/kW
Active power summer (april-sept)	41,90 NOK/kW
Consumption tax winter (jan-mars)	11,89 øre/kWh
Consumption tax summer (april-des)	16,44 øre/kWh
Consumption tax, reduced	0,58 øre/kWh
Energy fund fee	800 NOK/year

6.2.5 Incentive based demand response program

It is assumed that the EVPL is approved for offering flexibility to the local DSO as part of an IBDRP. The IBDRP is based on the *flexible consumption* IBDRP from the DSO Glitre Nett [6], which includes different product classes based on the notice time prior to activation. It is assumed that the EVPL is the only flexible source that can provide grid-service to the DSOs, while the office building remains a fixed load. The objective of this tariff is to provide grid stability services by reducing the EVPL demand when the DSO requests. By participating in this program, the office building receives economic incentives as a percentage reduction on the DCC. The incentive offers higher rewards for shorter notice times, as presented in Table 6.2.4.

When an IBDR signal is sent, it requests the EVPL to adjust the charging for a specified duration during the activation, and the EVPL may reduce the consumption depending on the requirements of the DSO. When V2G is included, the IBDR activation can provide load reduction in the range [-125, 125], meaning the EVPL can throttle to zero, and also deliver 125 kWh/h to the electricity grid or the office building which results in a lower total import for the system. This thesis assumes that the activation duration will always last for one hour.

Table 6.2.4: Flexible consumption IBDRP demand charge cost reduction by notice time

Notice time	**Demand Charge cost reduction in %**
0 hours	80 %
2 hours	50 %
12 hours	40 %

6.3 Algorithm input

6.3.1 Stochastic office building data

The stochasticity of the model is based on the uncertain office-building load. Only the SDDP-framework considers the stochastic data. Three scenarios are assumed for each decision stage, *low*, *medium*, and *high*. The *medium* scenario is equal to the historical office load data, presented in Section 6.2.1. *High* and *low* scenarios have a 30 % higher and lower office building demand respectively. The motivation behind the stochastic scenarios is to have a more realistic model with an uncertain future. The scenario percentages and probabilities for *high* and *low* scenarios are chosen quite arbitrarily and are not based on any previous work. The transition probabilities between each scenario are only based on which scenario is next. In other words, the origin scenario that is being transitioned from does not affect the transition probability. The probability of the *medium* scenario is 50%, *high* is 25%, and *low* is 25%. The cuts that are exported to be used in the short-term planning in RH are the *medium* scenario cuts. In some cases, deterministic cuts are used. These cuts are in practice generated by making the *high* and *low* scenarios equal to the *medium* scenario.

6.3.2 Input data to the RH algorithm

At the first day of operation, $P^{peak,init}$ is initialized to a value of 0. Further, implementing the IBDR-signals may result in the system not covering the daily aggregated EV demand. Because of this, a slack variable has been integrated into the system, introducing a cost of 30,000 NOK per Energy Not Served (ENS). This cost is not included in the total cost, as its purpose is only to ensure that the system does not optimize for any ENS, but is forced to deliver the requested charging if possible. However, in the economic analysis in this thesis, a penalty fee of 100 NOK/kWh for ENS is added to the total cost. The assumption for the size of this penalty fee is to visualize the economic impacts of not being able to provide the mandatory charging. Using a cost of 30,000 NOK/kWh would make it more difficult to compare the economic results.

6.3.2.1 Prediction horizons

The RH-framework uses the historical measured data for the office building load, which is equivalent to only using the *medium* scenarios. Thus, the RH-framework uses deterministic data which does not include any uncertainties. In the RH-framework, the length of the three horizons has to be determined. The SH considers the overall operational period of one month. The CH represents the hour of operation that is to be determined, resulting in a horizon of one hour. Lastly, the PH has a horizon equal to the time remaining of the operating day, thus the horizon will continuously decrease until the spot prices for the next day are revealed. The RH-framework has a new PH for each time step, where every PH receives a simplified representation of the rest of the SH from the linear cuts provided by the SDDP-framework. In

Norway, the spot prices for the next day are revealed at 1 PM [96], thus at 1 PM, the RH-framework will receive the spot prices for the next day. When this happens, the PH is expanded from 12 to 36 hours, revealing information about the next day of operation. When the new information is revealed, the PH will also receive new linear cuts from the SDDP-framework, which now presents a simplified representation of the cost of the remaining SH. The dynamic updating of the RH horizons is presented in 6.3.1, where the blue future cost curve is replaced with the red future cost curve considering the next day of operation.

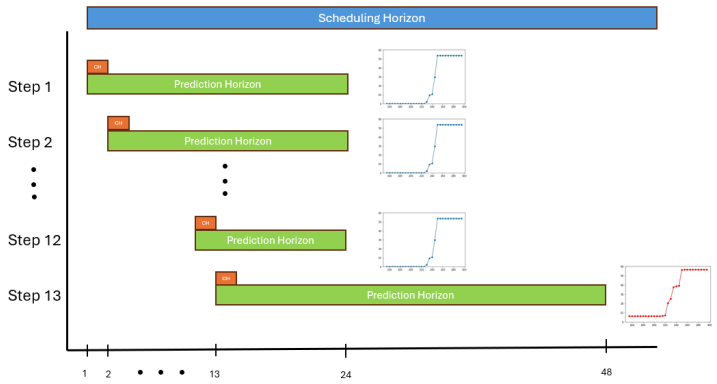


Figure 6.3.1: Visualizing the RH-framework with extended PH at 1 PM each day. Figure taken from **Paper I**.

6.4 Model Cases

The goal of Case 0 (C0) is to analyze the impact of the future cost curves provided by the SDDP-framework, and also establish an analysis that represents the system when no IBDR-signal is given. C0 will consider two months, namely the winter month March, and the summer month September. Three further case studies are developed, which are analyzed for for the month of March, each with its own goal and motivation. The common goal of the three cases is to analyze the consequences of IBDR-signals under different scenarios. Case 1 (C1) and Case 2 (C2) both analyze the consequences of IBDR-signals when no economic incentives are given to the office building, and rather look at the operational impact of the requested IBDR activations. In C2, V2G technology is implemented to facilitate a throttling signal that is greater than the maximum power capacity of the EVPL. Lastly, Case 3 (C3) includes economic incentives to see if an IBDRP can be of economic gain for the office building operator.

6.4.1 Case 0

C0 firstly presents an analysis of the SDDP-cuts. This is done to better understand how the future cost curves affect the operation of the model in later cases. The RH-framework is then tested with the future cost curves, to investigate if they have the expected impact on the short- and long-term operation. Additionally, an analysis of the impact of SDDP-cuts is performed, with both deterministic and stochastic linear cuts.

6.4.2 Case 1

C1 considers the IBDR-signal for the three notice times presented in 6.2.4. This case considers only the stochastic SDDP-cuts, and only the month of March is investigated. This work assumes that for every workday, the IBDR-signal is prompted with a given notice time to reduce EV charging at noon for 1 hour. The DCCR is not included, to better observe the consequences the different notice times have on the daily planning and operation of the EVPL.

6.4.3 Case 2

C2 is an extension of C1 by including the V2G technology, making the system able to send energy back to the grid and office building. This also considers that the IBDR-signal can make use of the V2G capacity, meaning full V2G-discharge is possible during the IBDR-signal.

6.4.4 Case 3

C3 introduces the DCCR as an economic incentive of participating in the IBDRP, as presented in Table 6.2.4. This case is an extension of C1, meaning that the V2G technology is not included. A analysis is performed on the operational and economical impacts of the IBDRP, to see if joining the already existing program is economically feasible for the office building operator.

Chapter 7

Results and Discussion

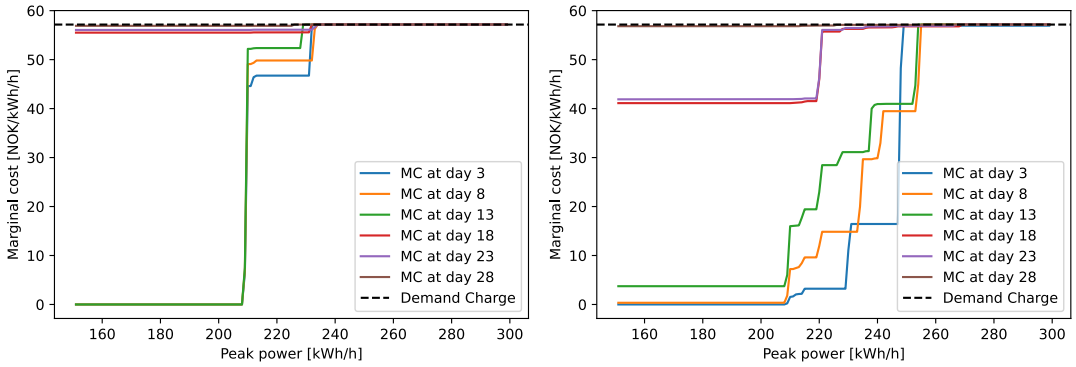
This section covers the results obtained through the simulation of the model, together with a discussion of the presented results. The important results of each case study are showcased. In the sections with economic analysis, only the operational costs are considered.

7.1 Case 0

In C0 the goal is to analyze the impact of the deterministic and stochastic future cost curves from the SDDP-framework. The future cost curves are first presented and analyzed to better understand how they impact the operation in the RH-framework. Lastly, four ways of implementing peak-cost management in the RH-framework are compared to investigate if the future cost curves are of value to the RH-framework.

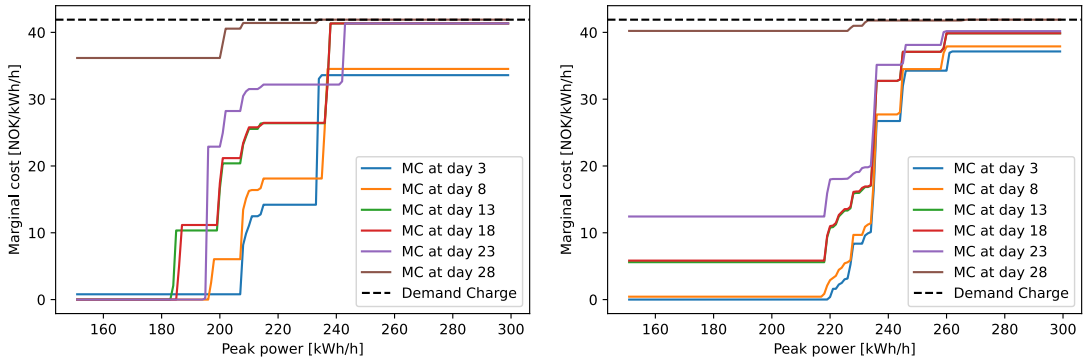
7.1.1 Future cost curves

In Figure 7.1.1 and 7.1.2 the marginal cost of increasing the peak power from the future cost cuts is presented. The curves are generated by solving α from equation (5.13) for a given day for initial peak values between 150 kW and 300 kW with a step length of 1 kWh/h. The curves represent the change in future cost, α bound by the strictest linear cuts for the respective peak value, as shown in Section 5.2.2. The curves are plotted for stochastic and deterministic cuts for March and September, together with the respective DCC for the month.



(a) Deterministic **(b)** Stochastic

Figure 7.1.1: Marginal cost of increasing peaks for March



(a) Deterministic **(b)** Stochastic

Figure 7.1.2: Marginal cost of increasing peaks for September

Plotted in Figure 7.1.2a are the future cost curves created for September using deterministic office building data. Observing day 3, the cost of increasing the peak value is seen to be zero until around 207 kWh/h. It is therefore cost-free to raise the peak, implying that the peak is going to be raised by a higher consumption-hour later in the month. At 207 kWh/h, there is a rapid increase in the marginal cost, implying that raising the peak further leads to a higher cost. In this area of the marginal cost curves present the short-term cost implication of raising the peak, which has long-term implications. The increase in the marginal cost is due to the DCC and the variation in spot prices. Increasing the peaks provide more opportunities for charging at lower spot prices, which helps decrease the marginal cost of increasing the peaks. From 235 kWh/h the marginal cost is constant for increasing peaks, where the cost is seen to be almost equal to the DCC.

Comparing the different days plotted in Figure 7.1.2a, it is observed that the later it is in the month, the more the curves are raised closer to the demand charge. This entails that the later it is in the month, the higher the cost is for increasing the peak value and the more likely it is for the model to not raise the peak. When plotting for the marginal future cost curves, it is expected that later in the month and the closer the model is to realizing the DCC, the higher the marginal cost should be for increasing the peak value. This is seen in all of the plots to a

varying degree. The later days have less potential to fully take advantage of the spot price variations. The weighting of the DCC and the cost of spot price are always competing. At the end of the month, the DCC has a bigger impact on the total cost than in the start of the month, because of fewer opportunities to take advantage of spot price variations.

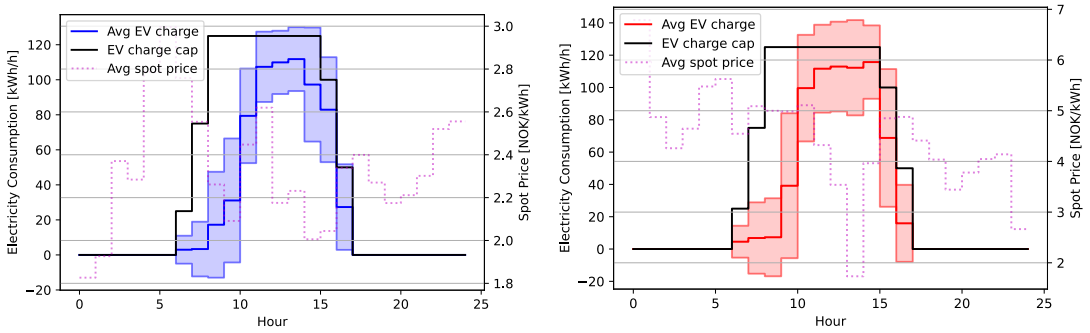
Further, a result of comparing the stochastic input data with the deterministic input data is seen on the curves with more detailed and smaller variations for increasing peak values caused by the stochastic load scenarios. These smaller bumps are cost levels which are because the model has to optimize for higher and lower loads than the "normal" load in the stochastic data. It can also be observed that the deterministic model generally has a higher marginal future cost for lower peak values, which indicates that the model values a lower peak and less flexibility in spot price variations. The stochastic model has to account for a possible increase in the office-building load of 30% and therefore values raising the peak.

SDDPs quality of only optimizing for the *interesting* state variable values is easily observed in the plots. In Figure 7.1.2b the marginal cost is seen to be constant at a lower value than the DCC for peak values above 260 kWh/h. This is an inaccuracy caused by the model never optimizing for peaks this high. The same phenomena can be seen for peak values lower than 215 kWh/h. Here, days 13 and later have a marginal cost higher than zero, which would not have been the case in reality. These days are not optimized and tested on these low peak values, and these values therefore lack detailed information on the cost of raising the peaks. The increased detail and accuracy in the peaks of interest decrease the detail and accuracy in the peaks outside of the researched range. Outer-lying peaks are represented poorly as they are not tested at all of the possible peak values. This could cause problems if something out of the ordinary (such as an IBDRP throttling signal) shifts the peak values outside of the researched range. Using a decomposition method that discretizes the state variable range into regular intervals such as SDP would increase the future-cost prediction accuracy for outer-lying peaks, but would also lead to inaccuracies between the discrete states used.

Figure 7.1.1a which shows the deterministic case in March, is an interesting case as the increase in marginal cost on a single peak value is very pronounced. The marginal cost is unchanged at 0 NOK/kWh/h until approximately 210 kWh/h for days 3, 8, and 13, entailing it is *free* to increase the peak until around 210 kWh/h. The later stages have a marginal cost almost equal to the DCC from the beginning. It is evident from this that the peak value for the month is "set" between days 13 and 18. A probable cause of this can be seen on day 16 in Figure 6.2.2a, where the office load has a high peak. A similar peak can also be observed on day 7, but since it is not the last peak, the marginal cost of increasing the peak to the predicted peak on day 16 is zero.

7.1.2 Operational impact of future cost curves

Figures 7.1.3 show the average EV charging with the respective average spot prices for four different months and the maximum EV charging capacity. These plots have used the stochastic future cost curves provided by SDDP.



(a) March 2022 **(b)** September 2022

Figure 7.1.3: Average EV charging with standard deviation and average spot price

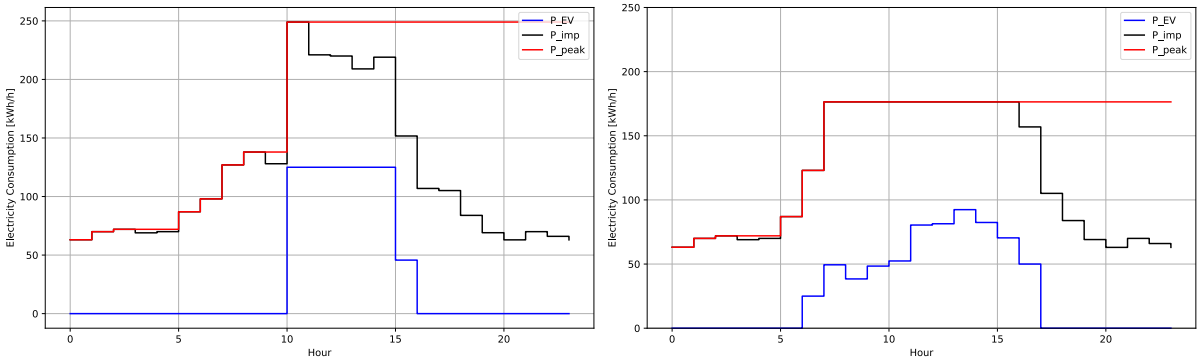
The figures show that the months have flexibility potential, meaning that it is possible to shift some of the EV-charging load. In these plots, the spot price and demand charge are weighted to create an optimal solution. An important observation is the model's preference to schedule EV charging later in the charging period and schedule almost no charging in the morning hours. This is due to the generally lower spot prices in the midday as seen in Figure 6.2.3 together with the lower office-building load in the midday, as seen in Figure 6.2.1.

7.1.3 Economical and operational performance with different peak-cost management

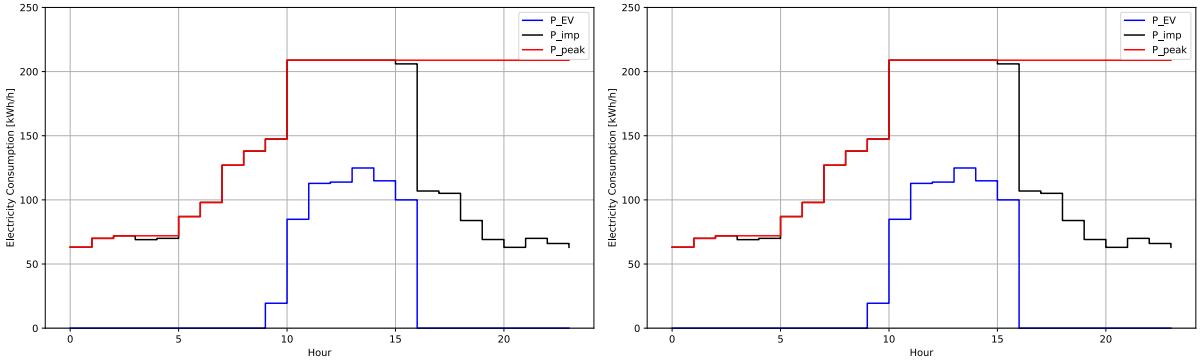
In this Section, the system performance will be analyzed for four different implementations of peak-cost management in the RH-framework. Here, the goal is to compare the efficiency of the SDDP-cuts for representing the peak cost towards other approaches. P_{no} represents the system when the future cost curves are not included, thus the system does not consider the DCC. P_{min} represents the system when it tries to minimize the peak each day, considering only the previously achieved peak. Lastly, the RH-framework includes the future cost curves with a deterministic and stochastic approach from the SDDP-framework, given as P_{det} and P_{stoch} respectively.

Table 7.1.1: Comparison of implementations of the demand charge

Month	**Cases**	P_{no}	P_{min}	P_{det}	P_{stoch}
March	Total cost [KNOK]	207.62	204.81	203.68	203.68
	Peak $[\frac{kWh}{h}]$	282.0	208.9	208.9	208.9
September	Total cost [KNOK]	360.41	360.84	359.11	359.22
	Peak $[\frac{kWh}{h}]$	259.9	196.5	208.9	222.3



(a) P_{no} **(b)** P_{min}



(c) P_{det} **(d)** P_{stoch}

Figure 7.1.4: First day of operation in March comparing implementations of the demand charge

In Figure 7.1.4, the operation in the first day of March is presented for the P_{no}, P_{min}, P_{det} and P_{stoch} approaches. The stochastic and deterministic future cost curves give the same result for March and are therefore shown in the same figure. The blue line is the power used for EV charging, the black line is the imported power from the grid and the red line is the peak power import.

When implementing P_{no} to the system, it does not consider the DCC, only the spot price, and it is therefore expected that this implementation will result in the highest monthly peak, which can be validated in Table 7.1.1. By not considering the resulting cost from the monthly peak, the DCC becomes high and provides an unnecessary high spot price flexibility potential. In Figure 7.1.4a it can be observed how the system utilized its full capacity between 10 and 15, resulting in a high peak.

For P_{min}, it is expected that the approach will have the lowest spot price flexibility compared to the rest of the implementations. This is because this implementation only has information on the past and the current PH regarding the highest peak, and therefore weighs having a lower peak when it would be beneficial to raise it. Figure 7.1.4b shows the first day of the charging pattern for P_{min}, and it can be observed that the peak is initially lower than the resulting monthly peak, evening out the peak load for almost all hours with charging.

Both P_{min}, P_{det}, and P_{stoch} manage to spread the load and minimize the peak. The difference between including the future cost curves is that the future cost curves incentivize the model to raise the peak earlier to account for spot price variations, as seen in the difference between Figure 7.1.4c and Figure 7.1.4b. Here, for P_{det} and P_{stoch}, the power peak is raised to the final power peak already on the first day. This results in an objective function cost saving due to spot price variations of 0.55% compared to P_{min}.

For September, P_{min} results in the highest total cost. Compared to P_{no}, this indicates that the peak is too low to maintain flexibility for later, resulting in a high energy cost. Having enough flexibility for spot price variation is even more important in September with greater fluctuations in spot price, as can be seen in the plots in Section 6.2.3. As September is defined as a summer month in the grid tariff, the demand charge is lower, which makes the system even more likely to increase the monthly peak.

Unlike March, P_{min}, P_{det}, and P_{stoch} do not have the same monthly peak, showing that the future cost curves incentivize raising the peak. Comparing the stochastic and deterministic future cost implementations, the stochastic approach is seen to value a higher peak by quite a substantial amount. This is probably due to the cuts generated on the *high* office load scenario. In Figure 7.1.2 the marginal cost of increasing the peak is plotted for September. This plot clearly shows how the stochastic approach has a near-zero marginal cost of raising the peak to almost 220 kWh/h, while the deterministic approach raises the marginal cost at approximately 200 kWh/h. The higher peak in the stochastic cuts only results in an increase in total cost of 110 NOK or 0.03%, which is a negligible amount.

Future cost curves in P_{det} and P_{stoch} are seen to incentivize raising the peak earlier, resulting in cost savings due to increased flexibility. In September, P_{min} results in the highest total cost due to insufficient flexibility, which shows the importance of flexibility regarding spot price variations. The stochastic SDDP-cuts for September, despite resulting in a higher peak, only marginally increase the total cost compared to the deterministic SDDP-cuts. P_{det} and P_{stoch} demonstrate the importance of SDDP-cuts for balancing the monthly peak and flexibility regarding spot price variations.

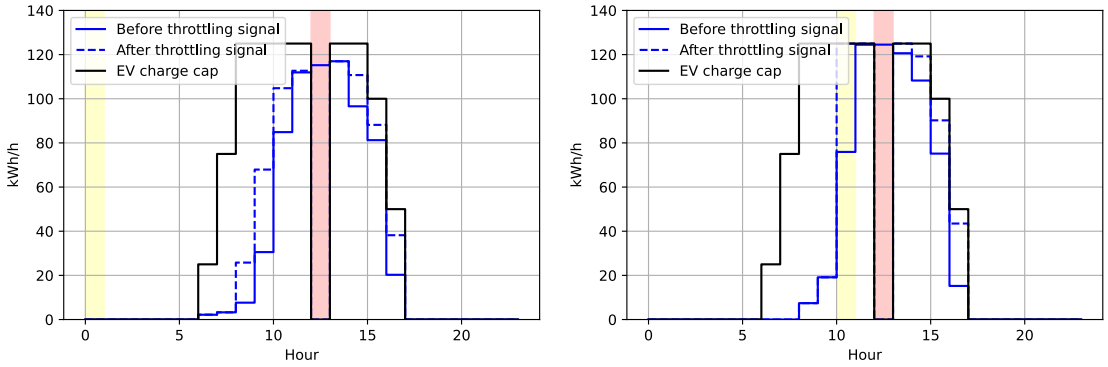
7.2 Case 1 - IBDRP

In Case 1 the goal is to analyze how the system reacts when an IBDR-signal is given for March. Based on the analysis done in Section 7.1, and hour 12 being a high load hour for the EVPL without the IBDR-signal as seen in Figure 7.1.3a, a throttling signal is given at hour 12 to analyze the flexibility potential and analyze how the RH-framework is proactive and reactive to sudden load changing signals. The system is also investigated for how the system performs under various notice times and increased throttling amounts. An economic incentive for throttling is not included in the optimization problem, to better observe the impacts made by the throttling signals. In all of the cases in this sections except 7.2.6, a worst-case scenario is explored. The worst-case scenario implies a maximum throttle signal every weekday, in the

entire month. In the marginal sensitivity analysis in Section 7.2.6, an IBDR signal is only given on the first day, to be able to analyze the long-term impact of one signal. In some of the analysis, the results are compared with March using the stochastic cuts. This is from here on known as the Base Case (BC).

7.2.1 Operational performance with IBDR-signal

In this section, a worst-case scenario is examined to analyze how IBDR-signals affect the systems operational performance. To visualize how the EV charging profile is affected by the IBDR-signals, Figure 7.2.1 presents the average EV charging before (full lines) and after (dotted lines) the signal is given. The maximum EV charging capacity is plotted to visualize how constrained the EV charging is when a throttling signal is given. The yellow bar represents the hour when the system is notified of the IBDR-signal, and the red bar represents the IBDR-signal activation hour.



(a) 12 hour notice time

(b) 2 hour notice time

Figure 7.2.1: Average EV charging in March before and after a maximum IBDR-signal given for every weekday

From Figure 7.2.1, it can be seen that there is limited flexibility when the notice time is 2 hours. The average charging is at full capacity at hour 11, giving the system only one hour before the IBDR-activation to shift to. Observing the *after throttling signal* line not being equal to the EV charging capacity in the 2 hour notice time, it can be deduced that the model does not charge at maximum capacity on all days. On some days with lower morning spot prices, the model chooses to allocate some charging in hours 8 and 9, which gives the model enough room later in the day to cope with the throttling signal. With hour 11 charging at max capacity, it is shown that the peak level is of little concern to the model.

Presented in Table 7.2.1 is the final peak value for each notice time, the amount of power that is shifted earlier $e^{shift}_{earlier}$ and later e^{shift}_{later} due to the throttling signal, the amount of power that was throttled e^{tot}_{thro}, and the amount of energy which was not able to be supplied to the EVs, ENS.

Table 7.2.1: Operational performance for the system under different notice times for a maximum throttling IBDR-signal for March

Notice time	***Peak***	$e^{shift}_{earlier}$	e^{shift}_{later}	e^{tot}_{thro}	**ENS**
BC	208.85	0	0	0	0
12	221.68	1638.9	896.9	2535.8	0
2	282	1141.19	1323.34	2748.85	373.87
0	259	0	1379.38	2828.95	1449.566

From the table it is evident that the system struggles to handle the throttled power in the worst-case scenario for shorter notice times. The 0 hour notice time case has a very predictable outcome, with only shifting power after the throttling signal, maxing the charging capacity. A reason for the system having lower flexibility potential for shorter notice times is the model's preference of allocating EV charging to later in the day, seen in Figure 7.1.3. This tendency is typically due to lower spot prices in the midday combined with a lower office-building load. This is shown to be a weakness of the model when the model pre-notice time signal has no information on when a signal can occur. Without any information, the model optimizes for the normal operation which mostly considers the peak cost and spot prices. Implementing a constraint that would force the system to have available flexibility at all times would likely increase the daily operational costs. Given that this scenario is a worst-case scenario, it is likely that the signals would occur more rarely and therefore the ENS would be lower. This problem is also a question of the cost and impact of not delivering the *promised* EV charging to the customers.

7.2.2 Peak increase due to IBDR-signals

For a worst-case scenario, the RH-framework struggles to accurately predict the final monthly peak, and therefore can not utilize the flexibility in spot price variations correctly. Figure 7.2.2 shows how the EV charging changes for three days when an IBDR-signal is given every day with a 12 hour notice time. The full lines represent the planned operation before the notice time of the IBDR-signal is given. The dashed lines represent the planned operation after the signal is revealed to the RH-framework.

As previously mentioned, the SDDP-framework does not consider the IBDR-signals, thus the future cost curves struggle to give the RH-framework an accurate estimation if the peak is raised too high by the IBDR-signals. When the PH in the RH-framework receives the IBDR-signal, it has to make a new plan for how to cover the load which was previously planned in the throttling hour. On day 6, when the IBDR-signal is given, the PH plans to solve the reduced EV charging capacity in the activation hour by being mostly proactive, shifting energy to before the throttle hour, which results in a monthly peak increase. The RH-framework does not know that the IBDR-signal will also occur the next day, which also results in a peak increase. The figure shows how the system is one day behind in raising the peak. If this new monthly peak was known for day 6, the system could potentially utilize this to have more flexibility regarding the spot prices.

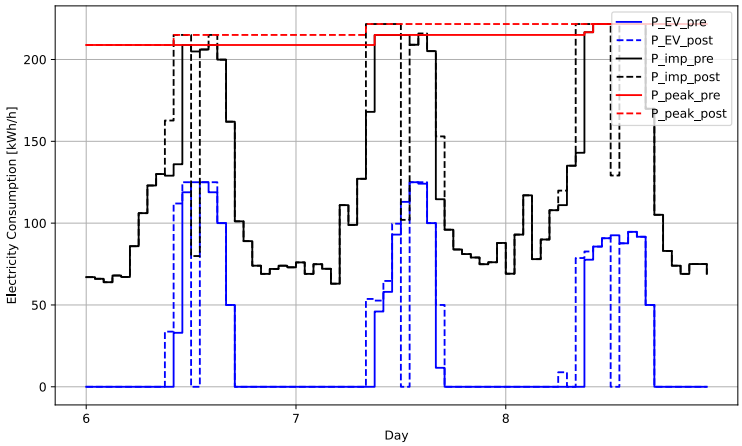
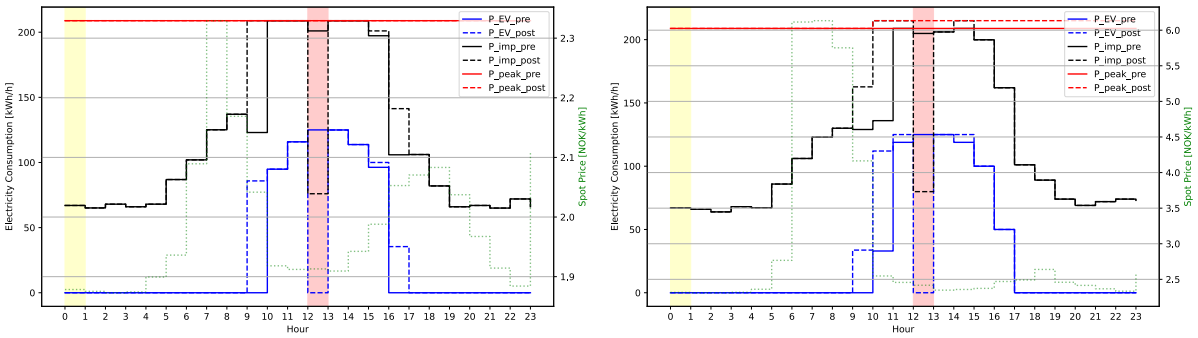


Figure 7.2.2: Change in operation for days 6, 7, and 8 with a 12 hour notice time for March

7.2.3 Daily operational performance with IBDR-signals

In this section, the daily operational performance during an IBDR-signal is investigated in detail for the worst-case scenario. The goal is to analyze how one day of operation reacts to an IBDR-signal, and how the system chooses to shift its energy to minimize the resulting total cost.

Figure 7.2.3 show the daily operational performance when the notice time is 12 hours before activation, where Figure 7.2.3a presents the load pattern for day 2. The figure indicates that the system has enough time and flexibility to shift the energy lost due to the activation hour so that the peak is unchanged. The model works proactively and reactively to avoid increasing the peak.



(a) Change in operation for day 2

(b) Change in operation for day 6

Figure 7.2.3: Comparison of changes in operation in March with a 12 hour notice time

Figure 7.2.3b presents operational day 6, where the monthly peak is increased. From the figure, it can be observed that the spot price is lower after the activation hour, but the energy has mostly shifted to earlier hours as the original charging strategy had mostly maximized the capacity in the afternoon. Based on Figure 6.2.4a, this will often happen, as the spot price is low in the afternoon. Because of this, the system is forced to act proactive by shifting energy before the activation hour. By shifting the energy to hour 10, the monthly peak increases. The

system could have charged at earlier hours but finds it optimal to increase the monthly peak rather than charging at high spot prices. In Section 7.2.2 it was mentioned that day 6 could potentially utilize knowing the next days peak-increase. From Figure 7.2.3b, it can be observed that this information would have been of value, as more charging could have been done in hour 10 rather than in hour 9.

Figure 7.2.4 shows the first day of March when the notice time is 2 hours. From this figure, it can be observed how every hour has maximized its capacity when the signal is given, resulting in a high increase for the monthly peak. However, the short signal time still results in an ENS of 1.3 kWh. This shows the importance of letting the system have significant notice time to avoid ENS. For this specific day, a 1 hour earlier notice time would have resulted in 0 ENS.

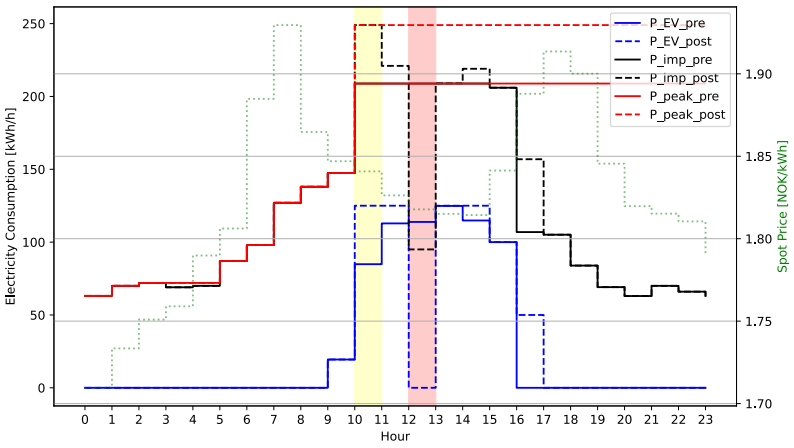


Figure 7.2.4: Change in operation for day 0 in March with a 2 hour notice time

7.2.4 Economic impact of the IBDRP

The goal of this section is to analyze the economical impact of the IBDRP for the the worst-case scenario. This analysis is presented in Table 7.2.2. The **Total Cost** includes all of the costs associated with the energy use in the building, being the grid tariff, spot price costs, and the cost of not supplying energy. The **Peak Cost** is the DCC from the grid tariff times the final monthly peak. **Shift Cost** is the difference in spot price costs from the shifting of allocated EV charging due to a throttling signal. **ENS Cost** is the ENS times the cost of not supplying energy, $C^{\text{not delivered}}$.

C1 has not included the incentives for participation in the IBDRP, thus the cost changes highlighted in this section reflect the cost of providing IBDR flexibility, which will then reflect the minimum profit that must be considered. Given that the cost of ENS is 100 NOK/kWh, the total cost is severely increased when the notice time is 0 or 2 hours. When the notice time is 12 hours, the resulting total cost is 0.57% higher for this worst-case scenario when considering that 23 IBDR-signals were activated during the month. This indicates that a quite conservative incentive for throttling can result in cost savings for the office building as long as the notice time is adequate. The Shift Cost for 2 hour and 0 hour notice times is not presented, as the ENS makes the Shift Cost negative. As the cost of not supplying energy is a

quite arbitrary number in this thesis, there might be other ways to handle the problem of not being able to fulfill the charging needs of the consumers.

Table 7.2.2: Economic performance for the system under different notice times for the IBDR-signal in March

Case	Notice Time	*Total Cost [kNOK]*	*Peak Cost [kNOK]*	*Shift Cost [kNOK]*	*ENS Cost [kNOK]*
BC					
	-	203.68	11.94	0	0
C1					
	12	204.84	12.67	0.49	0
	2	244.37	16.12	-	37.4
	0	347.89	14.80	-	144.95

7.2.5 Sensitivity analysis for increased throttling with an IBDR-signal given every weekday

This section will provide a more detailed analysis on the shifting of EV charging allocation on notice times of 2 hours and 12 hours, for reduced EV charging capacity in the activation hour ranging from 5 to 125 kW.

The 0 hour notice time is excluded due to the predictable behavior with the EV charging always being shifted after the throttling. The shifted power for an increased reduction in EV charging capacity in the throttling hour is plotted in 7.2.5 for 12 hour and 2 hour notice times. The blue bars indicate how much of the total shifted energy has shifted before the throttling hour, and the red bars represent energy shifted after the throttling hour, both given as a percentage of the total throttled power.

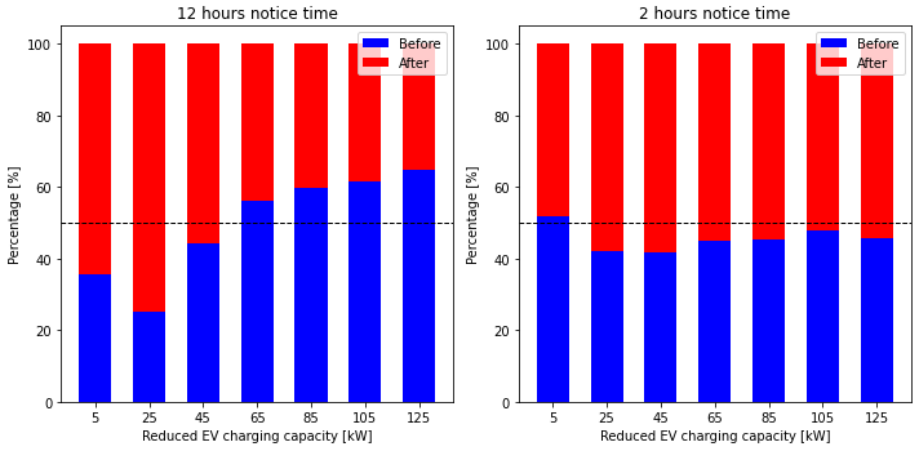


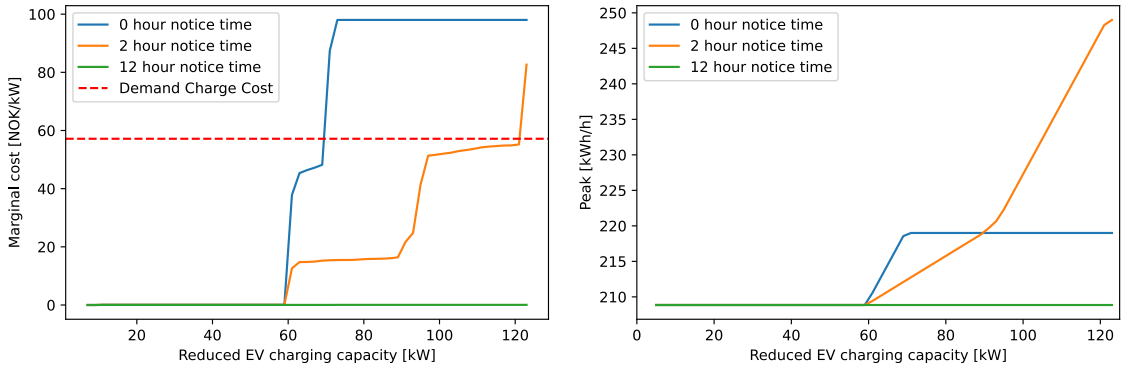
Figure 7.2.5: Shifted power for March when an IBDR-signal is given every weekday throughout the entire month

The figure shows that the 2 hour notice time scenario balances the allocation of the charging

before and after the throttling signal equally. The 12 hour notice time is shown to prefer allocating more EV charging after the signal for lower EV charging reduction signals, due to the lower spot prices later in the day. Variation in spot prices are so severe that the peak is even increased instead of charging in the highest spot price hours, as seen in Figure 7.2.3. With further increase in EV charging reduction, an increasing amount of the charging is moved to before the signal hour. This is the reason that the notice time of 2 hours in Table 7.2.1 is too constricted, as the signal happens at hour 10 which subsequently is too late to allocate EV charging to the morning hours.

7.2.6 Marginal cost analysis for increased throttling with an IBDR-signal at the first day of operation

A sensitivity analysis is performed on the reduciton of EV charging capacity. A signal is only given on the first day of operation. Figure 7.2.6a represents the marginal cost of reducing the EV charging capacity for the whole month of operation. Notice times of 0 hours, 2 hours, and 12 hours, are plotted as blue, orange, and green respectively. Figure 7.2.6b presents the corresponding monthly peak values. A cost of ENS, $C^{not\ delivered} = 100$ NOK/kWh, is added to the results after the RH-framework is done optimizing.



(a) Marginal cost

(b) Peak increase

Figure 7.2.6: Marginal cost of reducing EV charging capacity in hour of throttling for a notice time of 0, 2, and 12 hours. IBDR-signal given first day of operation in March.

In Figure 7.2.6a, it can be observed that when the notice time is given 12 hours before activation, an increasing throttling does not noticeably affect the marginal cost. The seemingly zero marginal cost of increasing the peak is in fact non-zero, but it is a lot lower than the two other cases. The 12 hour notice time is therefore plotted alone in Figure 7.2.3b. The marginal cost increase for a notice time of 12 hours is due to the difference in spot price from the hour of planned charging before the signal and after the signal. The peak stays at the same value for the whole range of reduced EV charging as observed in Figure 7.2.6b, indicating that the peak cost is quite strict. The highest marginal cost is found to be 0.047 NOK/kW which is when the reduced EV capacity is larger than 75 kW. These results show that when a 12 hour notice time is used, the cost of flexibility which could be of benefit to the DSO is very low.

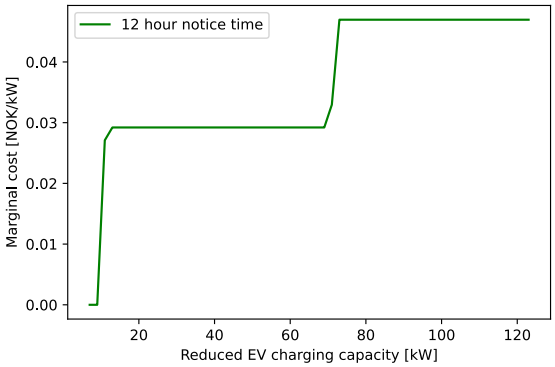


Figure 7.2.7: Marginal cost of reducing EV charging capacity in hour of throttling for a notice time of 12 hours. IBDR-signal given first day of operation in March.

The marginal cost increases significantly for both a notice time of 2 and 0 hours. There is a clear breaking point at around 60 kW reduced EV charging capacity, where the system can no longer avoid increasing the peak, as seen in Figure 7.2.6b, and the subsequent demand charge. The 2 hour notice time case quickly plateaus at a level around 15 NOK/kW, which suggests that the model has more than one hour to assign charging to. From Figure 7.2.6b, the peak can be seen to steadily rise 10 kWh/h in the interval when the reduced EV charging ranges from 60 to 90 kW, which results in a rate of increase on the peak value of around 0,33 kW per kW recused EV capacity. Including the DCC, this would result in a marginal cost of increasing the peak equal to $57.16/3 = 19.05$ NOK/kW. The difference between this marginal cost and the marginal cost seen in the plot at around 15 NOK/kW is the profit from increased flexibility to take advantage of spot price variation in the rest of the month. The 0 hour notice time only has one hour to allocate the peak-increase to, observed with marginal cost being close to the DCC. This indicates that two of the three hours used to distribute the peak increase in the 2 hour notice time case are before the notice time signal, which the 0 hour notice time case does not have "access" to when handling the IBDR-signal.

Continuing with the 2 hour notice time, when the reduced EV charging capacity reaches 90-95 kWh, two of the three hours that have been increasing their peak levels have reached their respective capacity and are "maxed out". Every kW increase in reduced EV charging now leads to a kW increase in the peak value, also observed in 7.2.6b. The marginal cost of increasing the peak is still not exactly equal to the DCC. This is also due to the raised peak ceiling providing more room to adjust for spot prices in the rest of the month. Lastly, the marginal cost can be seen to shoot up to 100 NOK/kW which is equal to the cost of ENS, $C^{\text{not delivered}}$.

To summarize, it can be seen that the notice time dramatically affects the marginal cost of increasing the reduced EV charging capacity. Being forced to increase the peak early in the month when you can not adequately plan for it can be costly. With the marginal cost for reducing EV charging capacity at 65 kW being 0.029 NOK/kW for a notice time of 12 hours, and 14,79 NOK/kW for a notice time of 2 hours, the importance of enough flexibility to account for the signals can not be understated! If a DSO was to request flexibility services from the office building/EVPL, the marginal cost curves are quite representative in the costs needed to cover the increased expenses for varying notice times and varying flexibility volumes

(kWh/h). Although this analysis is only performed for the month of March, further analysis has to be performed on other cases for a more general representation of the costs.

7.3 Case 2 - Including V2G

Like C1, the goal of C2 is to analyze how the system reacts when an IBDR-signal is given for the month of March for varying notice times. Now, the V2G technology enables the office building to satisfy an IBDR-signal from the grid that requires a reduction of up to 250 kWh/h. This section will analyze four sizes of IBDR-signals that require a reduction bigger than the maximum capacity of the EVPL. This section will also establish a V2G Base Case (V2G-BC). All of the cases in this section except 7.3.5, receive an IBDR-signal every weekday for the entire month. The analysis in this chapter is primarily done for a 12 notice time, except for the marginal cost analysis which is also performed with a 2 hour notice time.

7.3.1 Future cost curves

Figure 7.3.1 presents the marginal future cost curves which are used in the V2G case. No throttling signals are given as the SDDP-framework does not consider them. Many of the same observations that are seen in the no-V2G case in Section 7.1.1 are also applicable to these future cost curves. It is observed that the cost curves are most detailed around peak values of 220 to 250 kWh/h, which are the most common peaks seen by the SDDP-framework. The same inaccuracies are seen in the outer-lying points as in C0, with the peak values not being trained on them. A big increase in the marginal cost is seen between day 13 and day 18, created by the peak on day 16. A difference between these curves and the curves in Section 7.1.1 is that day 8 and day 13 start at a slightly higher marginal cost when V2G is included. Day 3 is also seen to have a slightly lower marginal cost for raising the peak with V2G available.

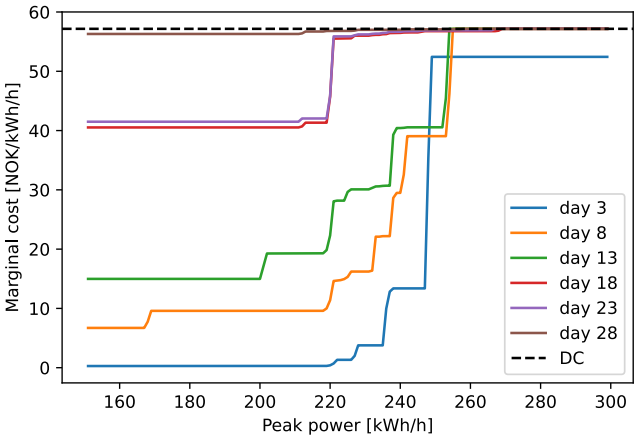
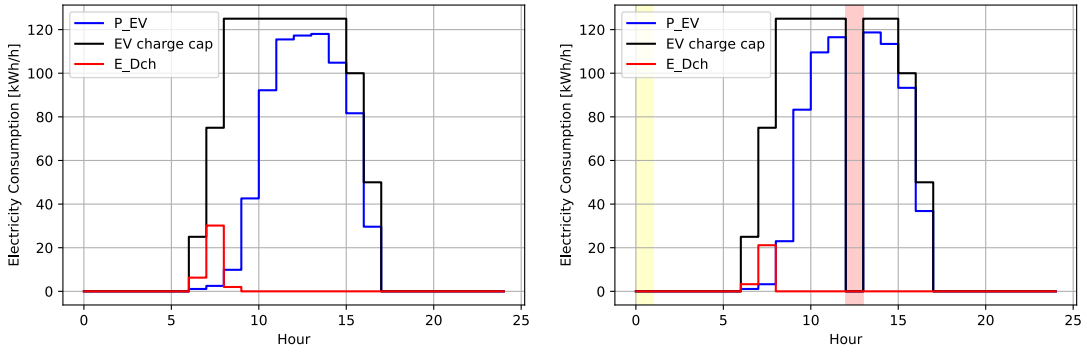


Figure 7.3.1: Marginal cost of increasing peak for March including V2G

7.3.2 Operational performance with a notice time of 12 hours

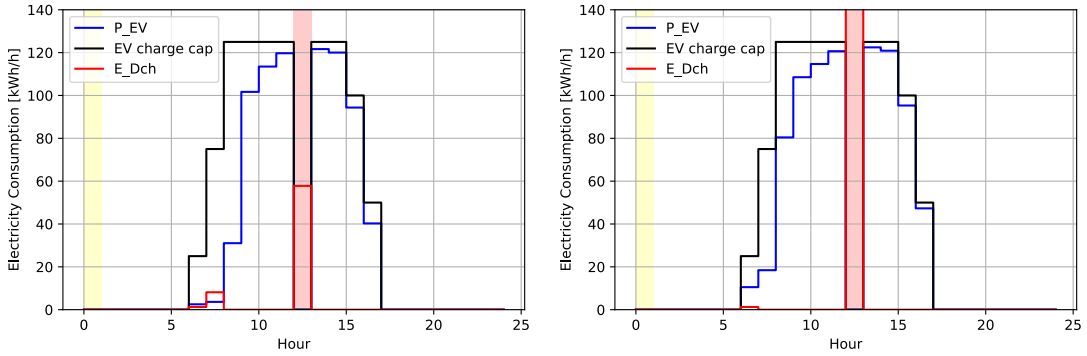
Figure 7.3.2a presents the average charging and discharging profile of the EVPL for the V2G-BC, where the system has the V2G technology implemented but does not receive any IBDR-signal. The system is seen to to discharge power from the EVs that arrive in the morning, due to the high average spot price in the morning as seen in Figure 6.2.3a. This energy has to be recharged later in the day. If the V2G technology enables the EVs to act as a BESS, first importing power, and then exporting power, an argument could be made that the V2G technology provides a greater value for the office building compared to the direct discharging of the EVs that arrive in the morning. Lastly, an observation is that the system still has the opportunity to respond both proactively and reactively, however, the flexibility for a reactive response is limited by the EV charging capacity.

Figure 7.3.2b visualizes the average EV charging and discharging when the reduced EV capacity and mandatory V2G power export is 0. Compared to the V2G-BC in Figure 7.3.2a, the system still utilizes the high spot prices in the morning to discharge but at a lower level due to the throttling in the activation hours. On average, the system has reacted proactively and reactively, but the reactive response is now even more limited.



(a) V2G Base Case **(b)** $P_t^{V2G,Thr} = 0$

Figure 7.3.2: Average EV charging with V2G



(a) $P_t^{V2G,Thr} = 50$ **(b)** $P_t^{V2G,Thr} = 125$

Figure 7.3.3: Average EV charging with V2G for throttling greater than maximum power capacity

Figures 7.3.3 present the average EV charging and discharging when $P_t^{V2G,Thr}$ is equal to 50 kWh/h and 125 kWh/h. $P_t^{V2G,Thr}$ represents the mandatory V2G export amount in the throttling hour. E_{Dch}, plotted as the red line, is the energy used in the discharging of the EVPL. E_{Dch} is seen to be higher than the mandatory discharge from the IBDR-signal because it has to account for the losses when discharging. Comparing the two figures, it can be observed that when V2G power export is 125 kWh/h, the model is more proactive, and the average discharging is reduced to almost 0. Figure 7.3.3b also shows that there is almost no remaining capacity for the system to be reactive. When the V2G power export is 125 kWh/h, the system must shift approximately 250 kWh (depending on the previously planned charging), which equals around 37% of the aggregated daily EV demand. Thus, the system has to utilize all available hours, including the peak hour, to cover the IBDR-signal and aggregated EV demand.

7.3.3 Operational performance with IBDR-signals

A more detailed analysis is performed on the system variables for the monthly operation when including mandatory V2G export for a notice time of 12 hours, shown in Table 7.3.1. The system is tested on mandatory V2G power export values of 0, 50, 120, and 125 kW. $e_{tot}^{EV,Thr}$ is the amount of energy displaced due to throttling. $e_{tot}^{V2G,Thr}$ represents the total amount of energy used by the EVPL to discharge due to the mandatory V2G export. e_{tot}^{Dch} represents the total amount of energy used by the EVPL to discharge for all purposes. e_{tot}^{Exp} is the exported energy to the grid, and $e_{tot}^{Dch,dif}$ is the difference in planned discharging before and after the notice signal is given.

The difference between $e_{tot}^{V2G,Thr}$ and e_{tot}^{Dch} is seen to decrease as $P_t^{V2G,Thr}$ increases. This shows how the system loses the flexibility to charge at the low spot prices and discharge at high spot prices when $P_t^{V2G,Thr}$ is high, as the system must prioritize to cover the lost energy due to the throttling hour. It can also be seen that the system has to start exporting energy to the grid when the whole office building load is covered by the V2G, as seen by the non-zero e_{tot}^{Exp}.

Table 7.3.1: Operational performance of the system under increasing mandatory V2G export using a 12 hour notice time

$P_t^{V2G,Thr}$	***Peak***	$e_{earlier}^{shift}$	e_{later}^{shift}	$e_{tot}^{EV,Thr}$	$e_{tot}^{V2G,Thr}$	e_{tot}^{Dch}	e_{tot}^{Exp}	$e_{tot}^{Dch,dif}$
V2G-BC	224.5	0	0	0	0	884	0	0
0	226.7	1706.9	626.3	2719.8	0	561.7	0	-334.4
50	237.2	2673.5	873.5	2796.4	1329.5	1545.5	1	649.3
100	239.8	3842.4	1054.2	2801.6	2659.0	2708.4	390.1	1812.2
125	244.5	4506.3	1140.7	2811.8	3323.7	3352.6	805.3	2452.5

From the table it can be observed that the shifted energy $e_{tot}^{Dch,dif}$ is equal to the energy throttled due to the reduced EV capacity, $e_{tot}^{EV,Thr}$, and $e_{tot}^{Dch,dif}$ including the losses. The losses due to charging efficiency must be included as the energy shifted must cover the lost energy due to the extra V2G charging. The balance between the shifted power and the energy requirement due to throttling can be presented as: $e_{earlier}^{shift} + e_{later}^{shift} = e_{tot}^{EV,Thr} + \frac{e_{tot}^{Dch,dif}}{\eta}$. When

$e_{tot}^{V2G,Thr}$ increases, the shifted energy must increase, which leads to a monthly peak increase. Both $e_{earlier}^{shift}$ and e_{later}^{shift} increase, showing that the system can act proactive and reactive. Given that most of the initial EV charging is planned to happen later in the day, there is a limit to how much energy can be moved to after the throttling signal. It is seen from the table that the system discharges 561.7 kWh from the EVPL when $P_t^{V2G,Thr}$ is 0 kWh/h, which is used to reduce the import during spot price hours.

7.3.4 Energy costs with increased mandatory V2G utilization

The economic performance for C2 given a 12 hour notice time is presented in Table 7.3.5. The reduced EV charging capacity is 125 kW in the analysis, while $P_t^{V2G,Thr}$ increases in four levels of 0, 50, 100, and 125 kW.

Table 7.3.2: Economic impact of increasing mandatory V2G export when with a 12 hour notice time

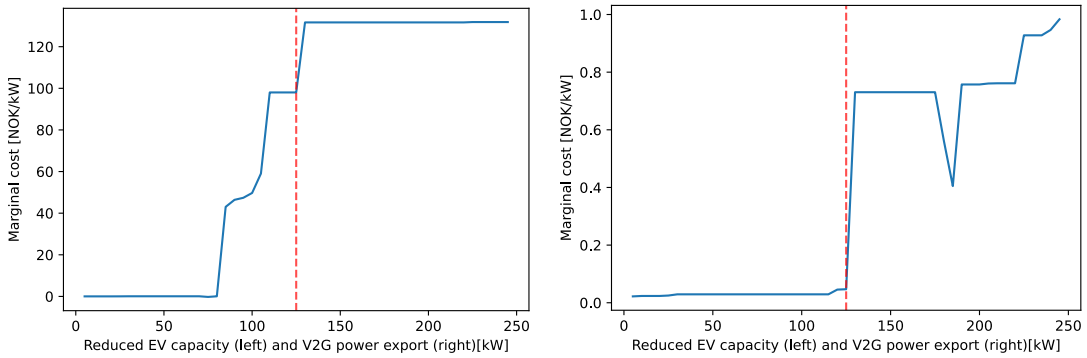
$P_t^{V2G,Thr}$	***Total cost [kNOK]***	***Peak Cost [kNOK]***	**Shift Cost *[kNOK]***
V2G-BC	203.57	12.83	0
0	204.71	12.96	0.94
50	206.77	13.32	2.44
100	208.89	13.70	4.41
125	210.19	13.73	5.47

When no IBDR-signals are given, the objective value for V2G-BC is 203.57 kNOK and the final peak value for the month is 224.47 kWh/h. Compared to the BC without V2G, this results in a reduction of total cost of 0.054%. The total discharged power from the EVPL in V2G-BC is 884 kWh. This indicates that the system utilizes the V2G technology, but the total cost savings are quite negligible. The reason for the low utilization of the V2G technology in V2G-BC is likely the charging and discharging losses which amount to a round-trip loss of 0.267 kW for every 1 kW that is charged and discharged. These losses significantly raise the marginal cost of using the technology, requiring a significant spot price difference for the technology to be profitable. Compared to C1, when the reduced EV capacity is 125 kW and $P_t^{V2G,Thr}$ is 0 kW, the V2G technology has lowered the total cost by 0.063%. However, both the peak cost and the shift cost increased, indicating that the system used V2G technology to export energy to the office building during peak hours, thereby minimizing the need for energy import.

For the increasing mandatory V2G export, a steady increase is observed in the total costs. Most of the cost increase is seen to be due to the spot price shift costs. The Peak Costs are seen to be increasing in a slower tempo, with $P_t^{V2G,Thr}$ at 125 kW only increasing the peak costs by 7% compared to V2G-BC.

7.3.5 Marginal cost analysis for increased throttling with an IBDR-signal on the first day of operation

Similar to C1 in Section 7.2.6, a sensitivity analysis is performed on the increased throttling, ranging from 0 to 250 kWh/h. Unlike C1, this section does not present the marginal costs for a notice time of 0 hour, as it will not manage to supply any V2G power export before reaching the point of ENS.



(a) 2 hour notice time

(b) 12 hour notice time

Figure 7.3.4: Marginal cost of increasing throttling with discharging from EVs using V2G for March

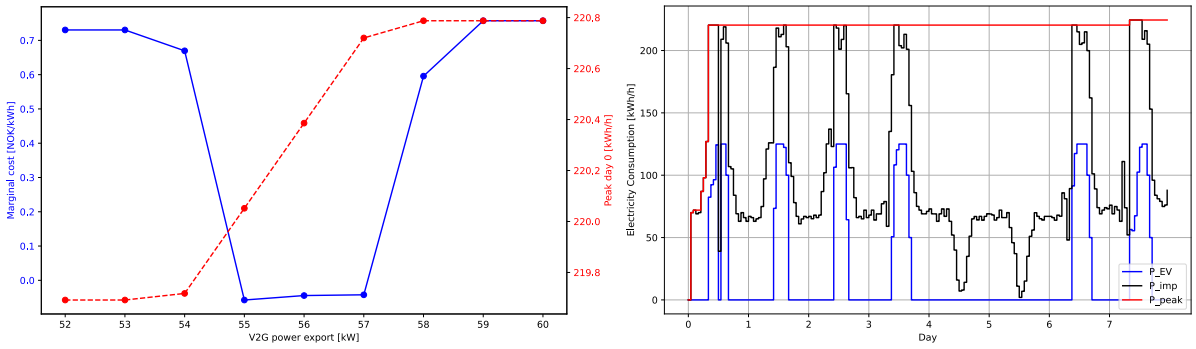
Figure 7.3.4a presents the marginal cost when the notice time is 2 hours. Compared to C1, the system now results in ENS even earlier. It can be observed that the marginal cost is constant when the EVPL exports power. This is because the system has ENS before the reduced EV capacity reaches 125 kWh. Any additional load, from V2G export or reduced EV capacity is not able to be covered by the EVPL. This entails that V2G has no value for the office building when the notice time is 2 hours, and the throttling demand requires discharging from the EVPL. The reason for the marginal cost being greater than the cost of not delivering power, $C^{\text{not delivered}} = 100$ NOK/kWh, is the losses included in the charging and discharging of power. The ENS variable is modeled as a slack variable, and therefore the ENS is thought to be imported and exported which includes the losses.

In Figure 7.3.4b it can be observed that ENS is avoided with a 12 hour notice time for the IBDR-signals. The marginal cost of increasing the reduced EV capacity is seen to be very low and quite constant at approximately 0.02 NOK/kW. This cost is all due to the spot price difference between the signal hour and the newly allocated charging. When the system has reduced the EV capacity to the maximum and starts using V2G, the marginal cost spikes to around 0.7 NOK/kW. This is because of the losses associated with both charging and discharging the EVs. At approximately 180 kW, the marginal cost has an unexpected dip. This dip is analyzed more thoroughly later through figure 7.3.5.

Further, the marginal cost has another increase for the 12 hour notice time when the V2G power export is 100 kW. This increase is when the V2G power export is larger than the office building demand, which is 95 kW in the activation hour. The exported power from the EVPL

that is not consumed by the office building is therefore exported to the grid. The exported energy to the grid does not include the volumetric cost in the grid tariff which has to be paid when importing power from the grid. The extra marginal cost due to the volumetric cost in the grid tariff is $C^{vol}_{jan-mars} = 0.1664$ NOK/kWh.

To further analyze the dip in the marginal cost, the marginal cost is plotted with a higher resolution of 1 kW in Figure 7.3.5a together with the peak power import on day 0. The marginal cost decreases and is negative for V2G power export values of 55, 56, and 57 kW, while the peak increases. This shows that when the peak increases on the first day, the total cost for the month improves. This is explained by using Figure 7.3.5b, which shows the hourly operation for day 0 to day 7 with a V2G power export of 56 kW. By increasing export on day 0, the peak is slightly adjusted, bringing it closer to the actual peak readjustment in day 7. The cost decrease with the increased spot price flexibility is not captured by the stochastic future cost cuts due to their stochastic nature. This is not the case with the deterministic cuts, as the marginal cost of increasing the V2G power export is strictly increasing with deterministic input data.



(a) Marginal cost for 52-60 kW V2G power export

(b) Operation from day 0 to day 7 with V2G power export equal to 56 kW in March

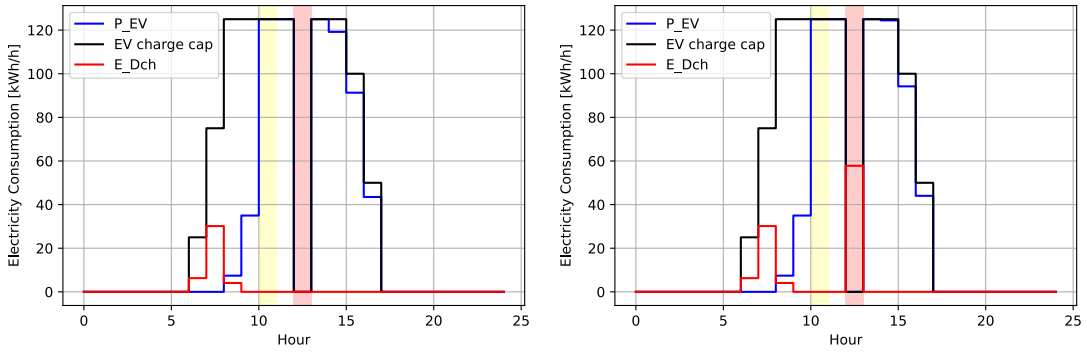
Figure 7.3.5: Marginal cost of increasing throttling with discharging from EVs using V2G

7.3.6 Challenges with V2G

V2G is a growing technology in grid flexibility, although the impact is not positive in all cases. To highlight some of the issues with integrating V2G in the modeled system, the 2-hour notice time case is investigated.

As seen from the marginal cost curves in Section 7.3.5, the 2 hour notice time leads to ENS if the IBDR-signal requires V2G power export during the activation hour. Figure 7.3.6a shows the average EV charging with EV power capacity and V2G power export at 0 kW in the activation hour. The same conditions apply for Figure 7.3.6b, except that the V2G power export is now 50 kW. The EV charging is either at, or close to the maximum EV charging capacity from hour 10 until the end of the charging period in both of the figures.

In C1, the amount of energy that could not be supplied was mostly dependent on how much EV charging was done before the notice time was given, but now it also depends on the amount of discharged energy. Comparing Figure 7.3.6a and Figure 7.3.6b, the discharging can be seen to be equal regardless of the size of the V2G power export in the activation hour. If an IBDR-signal is given after the system has chosen to discharge, the system can have problems reallocating the displaced energy from the throttling, leading to more ENS. In C1 with a notice time of 2 hours and reduced EV capacity of 125 kWh/h, ENS was 373.8 kWh. When V2G is integrated into the system, with the same notice time and EV capacity in the activation hour, the ENS is 1016.5 kWh, which is an increase of 271.9%. Thus, V2G power export early in the morning can be a big concern if there is an IBDR-signal notification after the discharging.



(a) $P_t^{V2G,Thr} = 0$ **(b)** $P_t^{V2G,Thr} = 50$

Figure 7.3.6: Average EV charging for March with V2G for throttling equal or greater than the maximum power capacity when the notice time is 2 hours

In Figure 7.3.7 day 3 is plotted for an IBDR-signal that is given every weekday, with a reduced EV capacity of 125 kW and $P_t^{V2G,Thr} = 0$. Due to the high spot price before the notice signal, the system plans to avoid charging and instead maximizes its capacity in the afternoon. The high spot prices also result in V2G energy export in the morning hours, minimizing imports during peak hours. The only difference between the planned charging before and after the signal is the lost power to the throttling hour. The throttling signal of 125 kWh results in an ENS of 125 kWh because the system has zero flexibility. Any additional mandatory V2G export in the activation hour on top of the reduced EV charging capacity is going to lead to more ENS and increased costs.

Battery degradation of the EVs has not been included in this thesis. As mentioned in Section 2.4.1.2, by integrating V2G technology, the battery's lifespan may be affected due to continuous charging and discharging. To motivate the EV owners to be a part of the IBDRP, and let the office building use the EV as a temporary BESS, some economic compensation is expected. It can be discussed if the savings of 0.054% would cover the potential battery-degradation compensation.

In this thesis, it is assumed that every arriving EV will be present in the EVPL for 8 hours. If this had been the case in the real world, a high degree of commitment by the EV owners would have been required. The commitment of the EV owners is crucial for two things. Firstly, when

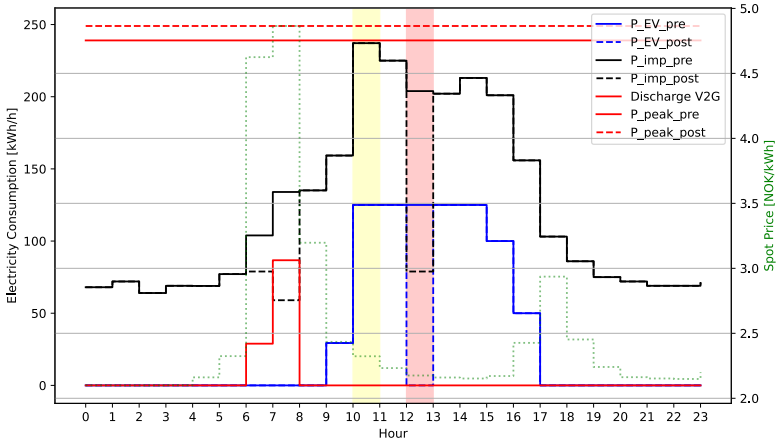


Figure 7.3.7: Change in operation for day 3 in March with $P_t^{V2G,Thr} = 0$ and maximum reduced EV capacity

discharging the EVs that arrive early in the morning, the EV owners must commit to staying in the EVPL if they want to ensure complete charging. Secondly, by being a part of an IBDRP where the office building can throttle more than the maximum capacity, the EVs must be present in the EVPL if an IBDR-signal is given. That is, there has to be a number of cars present during the activation hour for the office building to be able to prove V2G export services. This in turn makes the available hours of which the system can provide a V2G demand response service quite low, restricted to work hours on weekdays.

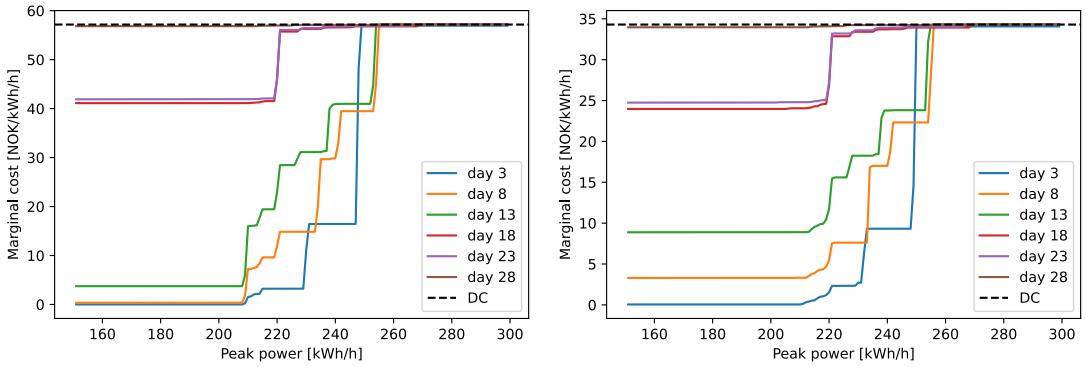
7.4 Case 3- Demand Charge Cost Reduction

In C3 the system is analyzed with an IBDRP which includes economic incentives. Similar to C1 and C2, the IBDR-signal is activated at hour 12, and the same notice times of 0, 2, and 12 hours are used. The three notice times entail three different levels of DCCR in the IBDRP, presented in Table 6.2.4. Through this analysis, the goal is to find if the economic incentives for participating in the existing IBDRP are of value to the office building, and if the DCCR covers the extra costs associated with the participation in the IBDRP. The impact of the reduced DCC on the systems operational planning is analyzed. V2G is not included in this section for two reasons, as V2G is shown in Section 7.3.6 to further constrict the system, and the IBDRP in Table 6.2.4 does not include exported power.

7.4.1 Future cost curves

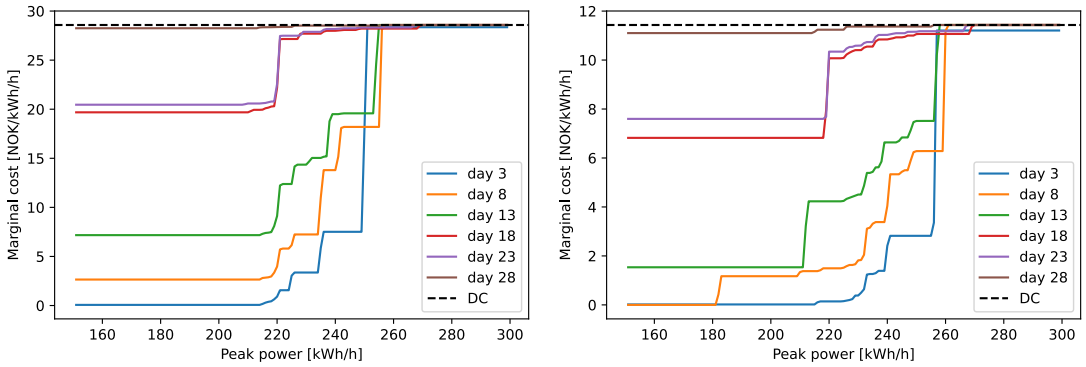
In this section, the future cost curves are analyzed to see how they are affected by the different DCCRs, and how they differ compared to C1. No throttling signals are given as the SDDP-framework does not consider them. Figure 7.4.1a is a repeated presentation of the future cost curves for March presented in Section 7.1.1, for comparison purposes. Figure 7.4.1b, 7.4.2a and 7.4.2b presents the marginal future cost curves with their respective DCCRs.

The cost curves have very similar shapes to what is seen in C1, and in Figure 7.4.1a. It is important to notice the different y-axis that the curves are plotted on. Further, the peak value interval at which the cuts are more detailed has the same starting point at approximately 210 kWh/h. The cost curves with a higher DCCR are more detailed and are lower for higher peak values. This is as expected, as a lower cost of the final peak value is going to incentivize raising the final peak value because of the weighting between spot price flexibility and peak costs. The future cost curves having similar shapes for the different DCCRs show that the office-building load and spot prices are controlling of the operation of the EV charging. Further, the DCC is also seen to be the controlling factor of the maximum marginal cost.



(a) 0% reduction **(b)** 40% reduction

Figure 7.4.1: Marginal cost of increasing peak for March using a DCCR



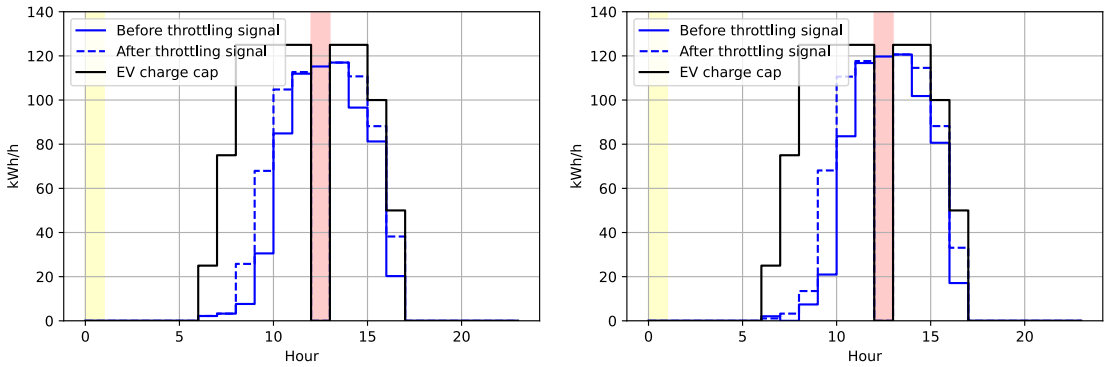
(a) 50% reduction **(b)** 80% reduction

Figure 7.4.2: Marginal cost of increasing peak for March using a DCCR

7.4.2 Change in the planning of EV charging due to a demand charge cost reduction

The purpose of this section is to visualize how the model changes the operation of the EVPL assuming a worst-case scenario for the IBDR-signals, including DCCR. To highlight the changes, the results are compared with the results in C1. Figure 7.4.3 presents the average EV charging for C1 and C3. The figures show that the EV charging pattern is very similar, and

the displaced energy due to throttling utilizes the same hours throughout the day. The primary difference between the two cases is caused by the increased monthly peak.



(a) 12 hour notice time from C1

(b) 12 hour notice time from C3

Figure 7.4.3: Average EV charging in March before and after an IBDR-signal, comparison of C1 and C3 with the worst-case scenario

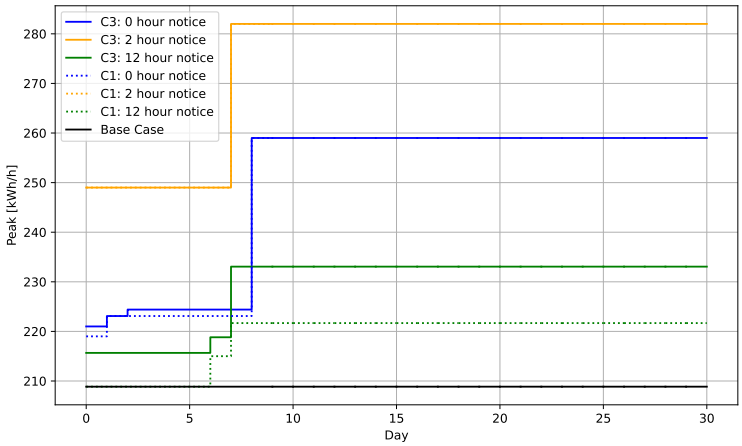
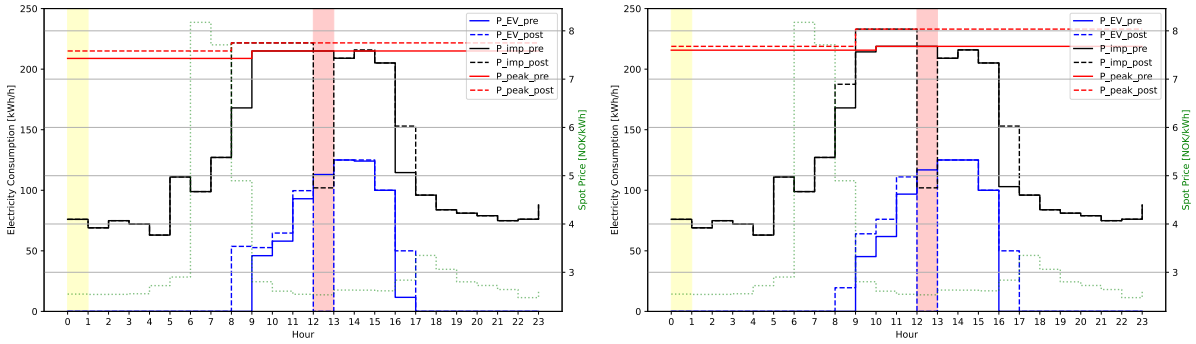


Figure 7.4.4: Daily peaks for March in C1 and C3 with worst-case scenario

Figure 7.4.4 presents the peak value development through the month for C1 and C3. It can be seen how the IBDRP with low notice times is compensated through the reduction in DCCR and subsequent peak costs. This also works the other way, as the model is now incentivized to raise the peak because of the lower marginal cost. The 12 hour notice time case is seen to be the only case which has the flexibility to weigh the demand charge and spot price variations. The C3 12 hour notice time peak is approximately 10 kWh/h lower for the entire month. The 0 hour notice time and 2 hour notice time peaks are almost exactly equal with and without the DCCR, indicating that the peak is forced and not chosen. Further, days 7 and 8 are critical for the monthly peak value, as these are the days that significantly raise the peaks. The difference in peak planning in the base case versus the cases with throttling signals is also evident, with the model not quite being able to optimize and plan for the increased peaks caused by the IBDR-signal. Optimally, the peaks should be set on day 1 to allow for the most flexibility in spot price variations if necessary.

Lastly, Figure 7.4.5 presents how the peak increases at day 7 for C1 and C3. The increased monthly peak in C3 enables the system to allocate more charging at lower spot price hours. Both cases react proactive and reactive, however, in C3, the energy shifted before activation can be distributed over several hours where the spot price is low due to the peak increase. At the same time, the system in C3 could increase the peak further to avoid charging at hour 8 but find it more optimal to have a lower peak.



(a) Change in operation *C1*

(b) Change in operation *C3*

Figure 7.4.5: Comparison of changes in operation for day 7 for *C1* and *C3* for March with worst-case scenario

7.4.3 Operational performance with IBDR-signals and a demand charge cost reduction

This section analyzes how the incentives in the IBDRP change the operational performance of the system for the worst-case scenario. Table 7.4.1 presents the operational performance for the 3 notice times for C1 and C3, both having using the worst-case scenario. BC is included for peak comparison.

Table 7.4.1: Operational performance of the system with different notice times in March

Case	Notice time	***Peak***	$e^{shift}_{earlier}$	e^{shift}_{later}	e^{tot}_{thro}	**ENS**
BC						
	-	208.85	0	0	0	0
C1						
	12	221.68	1638.9	896.9	2535.8	0
	2	282	1141.19	1323.34	2748.85	373.87
	0	259	0	1379.38	2828.95	1449.57
C3						
	12	233	1918.46	835	2753.5	0
	2	282	1123.6	1355.8	2872.6	393.3
	0	259	0	1388.7	2842.7	1454

From Table 7.4.1, the monthly peak only increases when the notice time is 12 hours when comparing to C1. The monthly peak does not increase when the notice time is 2 or 0 because the system reaches ENS, leaving no additional capacity for EV charging, which prevents further peak increase. The ENS is seen to be higher in C3 for notice times of 2 hours and 0 hours. Observing the daily peaks when the notice time is 2, it is found that C1 has an initial peak equal to 208.8 kWh/h and C3 has 217.6 kWh/h. This results in a different planned EV charging strategy before the signal is given, changing the amount of ENS.

In Table 7.4.1, the monthly peak has increased compared to C1. As the IBDRP for a 12 hour notice time provides a 40 % reduction in the DCC, which reduces the marginal cost of raising the peak, an increase in peak is expected. The weighting between the ability to adjust for spot price variation and the DCC is therefore shifted. Raising the peak value leads to more charging in cheaper spot price hours, which often also leads to more charging planned in the hour of throttling activation. If the system has a higher planned charging in the potential activation hour, and an IBDR-signal is given, this will require more shifted energy.

7.4.4 The potential of 0 and 2 hour notice times

Until now, only an IBDR-signal with the worst-case scenario has been analyzed for C3, which has led to the 0 and 2 hour notice times having ENS. The goal of this section is not to find the optimal IBDRP when the notice time is 0 or 2 hours, but to analyze the potential of these notice times without the worst-case scenario.

It is found that when the notice time is 0 hours, even an IBDR-signal that requires 1 kW EV capacity reduction every weekday leads to ENS. However, there is only one day in the month with ENS, as this is the only day which maximizes maximizes the EV charging capacity in the hours after the activation. If this critical day does not receive an IBDR-signal, it can be expected that a larger EV capacity reduction can be applied in the activation hour without resulting in ENS. However, it is found that when the EV capacity is reduced by 5 kW, there is ENS for days 7, 18, 28, and 31. There is shown to be limited flexibility in the system for a notice time of 0 hours as it struggles to avoid ENS. At the same time, it is seen from Figure 7.4.4 that if an IBDR-signal is not given on day 7, the peak does not increase, resulting in a changed planned EV charging. This entails that it is difficult to deduce which days of the month which are critical for the operation to avoid ENS.

For notice time 2, the system does not lead to ENS if the IBDR-signal, given every weekday, requires an EV capacity reduction of 104 kW or less during the activation hour. However, 0 ENS can not be guaranteed due to the potential changes in planned EV charging when the IBDR-signal is not given every weekday. Without IBDR-signals, the monthly peak is 217.66 kWh/h due to the 50% reduction in DCC that the 2 hour notice time provides. If an IBDR-signal is given on any day other than day 0, it is uncertain whether the planned EV charging has the flexibility to shift 104 kW to the other hours. If only one day in the month is seen to be critical for ENS, the system could risk entering the IBDRP and hope for no IBDR activations. Further, it is found that when the reduced EV capacity is bigger than 104 kW,

ENS occurs on multiple days. The reduction in EV capacity equal to 105 kW results in 19 days having the same amount of ENS, which also indicates that the planned EV charging is quite similar across the month.

From the analysis, it can be discussed if an IBDR-signal with notice times of 0 hours and 2 hours can be of potential benefit for the office building. If the IBDR-signal is received only once a month, it is possible that notice times of 0 and 2 hours could yield 0 ENS, however, this depends on three things. First, which days the IBDR-signal is given. Second, the planned EV charging for the given day. And last, the amount of reduced EV capacity in the activation hour.

7.4.5 Economic impact of the IBDRP with a demand charge cost reduction

The economic performance of the system when the IBDRP entails a reduction in the DCC is investigated in this section, using the worst-case scenario. The reduction of the monthly DCC is the same regardless of how many signals are given from the grid operators. Table 7.4.2 presents the energy costs for the 3 notice times for C3, C1, and BC. C1 and BC are included for comparison purposes.

Table 7.4.2: Economic performance of the system for varying notice times in March

Case	**Notice Time**	***Total Cost [kNOK]***	***Peak Cost [kNOK]***	***Shift Cost [kNOK]***	***ENS Cost [kNOK]***
BC					
	-	203.68	11.94	0	0
C1					
	12	204.84	12.67	0.49	0
	2	244.37	16.12	-	37.40
	0	347.89	14.80	-	144.95
C3					
	12	200.00	7.99	0.38	0
	2	238.20	8.06	-	39.33
	0	336.48	2.96	-	145.40

In C3, when the notice time is 0 or 2, the total cost for the system increases significantly compared to when the notice time is 12 hours. The reason for this is due to the increased ENS. However, it can be observed that the Peak Cost has been significantly reduced for all the notice times compared to C1. This reduction results in a Total Cost that is lower for all the notice times when compared to C1. When the IBDR-signal is given 12 hours before activation in C3, the Total Cost is reduced by 1.8 % compared to BC. This showcases how the IBDRP can be profitable for the office building for the 12-hour notice time program, even though C3 represents the worst-case scenario. If the IBDR-signal is given more rarely, or not at all, the total cost saving would be even higher in C3.

Comparing the 12 hour notice time between C1 and C3, the DCCR results in a total cost saving of 2.36%. It is seen that the Peak Cost decreases significantly. In Section 7.4.3 it was shown that C3 had a higher planned charging in the potential activation hour, and it was expected that this could lead to a higher shift cost. However, from the table, it can be observed that C3 has a lower Shift Cost than C1. The increased flexibility due to the increased peak has led to a decrease in shift cost from being able to charge more at low prices, even though more energy is displaced.

7.4.6 Marginal cost analysis for increased throttling with an IBDR-signal for the first day of operation

A sensitivity analysis is performed on the marginal cost of increased throttling, with an IBDR-signal only given for the first day of operation. The notice time for the IBDR-signal is coupled with its respective DCCR. The marginal cost is plotted in Figure 7.4.6.

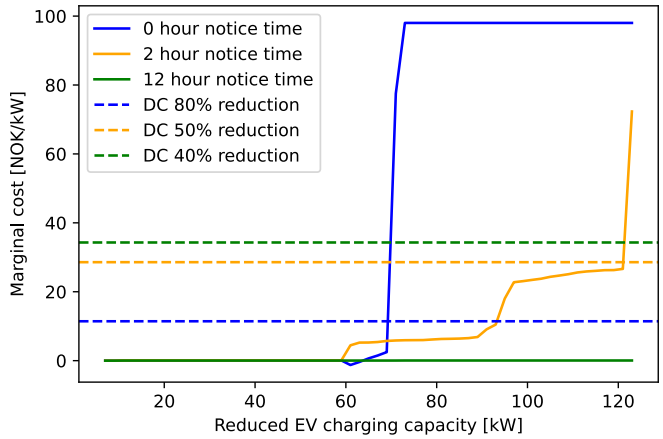


Figure 7.4.6: Marginal cost in March for reduced EV charging capacity in hour of throttling for a notice time of 0, 2, and 12 hours, with DCCR

Comparing the figure with the marginal cost without DCCR in Figure 7.2.6a, the 12 hour notice time scenario is quite similar. This is expected as the marginal cost was low and quite constant from before, and introducing a reduction in the peak costs would only lower the marginal costs. A more interesting case is the 2 hour notice time. The shape of the curve is quite consistent with and without a DCCR. However, the 50% reduction is seen to lower the curve by approximately. This observation shows that most of the marginal cost of reducing the EV charging capacity above 60 kW is due to peak costs.

Analyzing the 0 hour notice time case, a dip in marginal cost is seen when the first peak values are seen to impact the marginal cost at around 60 kW. This dip is caused by the same factors as mentioned in Section 7.3.5. The "forced" increase of peak because of the throttling signal around 60 kW resulted in a small profit. This is because of the increased flexibility in spot price variation and that the peak is shifted to be closer to the actual highest peak later in the month. This peak increase is deemed too "risky" by the SDDP-framework because of the possibility of lower and higher scenarios and therefore is not exploited by the model. After the

dip, the marginal cost is seen to increase from 0 to 100 NOK/kW, caused by the cost of ENS, $C^{\text{not delivered}}$. The increase in the 0 hour notice time marginal cost is even faster than without a DCCR, indicating the system is fine with raising the peak until the EV charging demand can not be met.

7.5 General Discussion

There are many assumptions made in the modeling of the system. To discuss how they affect the results, and if any assumptions should have been altered, a general discussion chapter is provided.

In this thesis, an IBDR-signal has in most analyses been given on the first day of operation, or every weekday throughout the entire month. An IBDR-signal every weekday, which in this thesis is considered a worst-case scenario, is analyzed to see if the system can handle the maximum possible strain.

If an IBDR-signal never occurs, participating in the program would provide direct cost savings for the office building. However, this would only lead to a cost increase for the grid operators, but at the same time, the DSO will have insurance on how they can resolve peak power problems in the grid. With the grid becoming more constrained, peak power problems in the grid can increase, which can also lead to situations where more IBDR-signals are sent.

Bi-directional chargers can have a higher price due to the additional components needed to manage the two-way flow of electricity. This thesis found that the cost savings from V2G was limited. It can therefore be discussed if the cost savings will cover the potential additional cost of the bi-directional chargers and the degradation of the EV batteries. If another month had been analyzed, where the spot prizes had even more variation throughout the day, it could have been found that the V2G technology was more valuable for the system. However, a positive with the V2G technology is that it enables the system to meet an IBDR-signal which requires more power than the EVPL can throttle in the activation hour. The possible economic benefits of providing extra power to the grid were not discussed and could have helped the V2G technology become more profitable. It was also assumed that all the EVs that used the EVPL were capable of bi-directional charging. Today this is an unlikely case, however, the V2G technology is under development, and in the future, more cars could be capable of bi-directional charging.

The energy not supplied variable in the model is quite an important factor when the costs were discussed. In the model it was assumed that the EVPL demand should be met, no matter the cost. It is seen that the ENS makes the 0 hour and 2 hour notice times difficult to operate. Further, the cost of the energy demand that could not be met is dependent on the agreement with the EVPL owner and the EV owners. An agreement could be possible that states if there was to come an IBDR-signal, the EVPL can not be held accountable for the lost charging. Or the EV owners can be asked to charge for a longer time if they want a full charge. It is

probably possible to come to an agreement where the cost of ENS is not as detrimental to the total operational costs as the $C^{\text{not delivered}} = 100$ NOK/kWh/h.

The modeling of the EVPL required the aggregation of the EV data. This was done due to the lack of information available on the individual operation of the EVs. A detailed individual modeling of the EVs could provide a more detailed insight into the operation of the EVPL. The participation in the flexible charging and V2G technology could be optional for the EV owners, reducing the flexibility potential. There could also be individual data available on the charging times, charging power, arrival and departure times, battery SOC at arrival and departure, battery charging profile, losses, and so forth. These input values would either require empirical data or many assumptions, which could worsen the results in the same way that aggregating the EVs does. Individual modeling of the EVs would increase the computational effort of both the real-time modeling and the creation of the future cost curves.

SDDP is a modeling framework that can handle multiple state variables without the computational efforts being too extreme. The model developed in this thesis only has one state variable, indicating that other decomposition methods could have been used. SDP is another decomposition method that discretizes the state variable, mentioned in Section 4.2.2 of the Theory. SDP was used in [58] to represent the long-term implications of the demand charge. The future cost curves presented in the study using SDP were more detailed in the outer-lying areas of the possible values for the state variable. This is caused by the discretizing of the state variable not weighting the "most interesting" values as in SDDP. This could implicate that the SDP algorithm would be better suited when optimizing a system for handling unexpected IBDR-signals. Throughout the results, the IBDR-signals are seen to raise the peak values above what was seen in the BC. This is seen to sometimes bring the peak values outside of the detailed range when estimating the future cost implications of the demand charge, which could cause inaccuracies.

The hour of the day which is throttled was chosen to be hour 12. This choice is based on hour 12 being a high load hour for the EVPL without the IBDR-signal in March, in addition to giving the system the potential of being proactive and reactive. With the spot price in March being generally high in the morning, the model is seen to often allocate the EV charging to later in the day when the spot price is lower, reducing the amount of energy that can be moved reactively. If the activaiton hour was at hour 11, the cases with notice time of 0 hours and 2 hours would have more hours to choose from when reallocating of displaced energy from the throttling. Potentially, this could have reduced the amount of ENS in when using a notice time of 0 and 2 hours.

A more realistic implementation of the throttling activation hour is to factor in when the grid has the highest power demand. This will vary, but as the spot price is set by the market through demand and supply, the grid is expected to be the most stressed during high spot price hours. If the activation hour is in the morning, the cases with a notice time of 0 hours and 2 hours would have greater flexibility in the reallocating of energy. At the same time, an activation hour early in the morning would entail a low amount of energy throttled. The system would be forced to act mostly reactive and the signal would be easily handled. In

reality, there is no restriction that the activation can only be given one time during a day. More throttling hours throughout the day would of course only entail a more constricted system and less flexibility.

Chapter 8

Conclusion

This thesis investigated how an electric vehicle parking lot (EVPL) can use flexible resources to reduce the cost of operation for an office building while providing grid stability services through an incentive-based demand response program (IBDRP). In the thesis a model is developed using the decomposition method Stochastic Dual Dynamic Programming (SDDP) and the real-time operation framework of Receding Horizon (RH). The SDDP-framework creates future cost curves which represent the future cost of increasing the peak demand. The future cost curves are seen to be accurate in the normal operating interval for the peak values, but can encounter inaccuracies in the outer-lying peak values. The RH-framework is implemented so that the operation of the EVPL can be optimized while simultaneously handling sudden IBDRP-signals.

The model is tested for the given system with IBDRP-signals using notice times of 0, 2 and 12 hours. In the worst-case scenario with maximum throttling at hour 12 every weekday, the 12 hour notice time is proven to handle the unexpected signal quite easily, with only a 0.57% cost increase compared to BC. It was shown through the sensitivity analysis that the marginal cost of providing grid services through an IBDR-signal was 0,047 NOK/kW at the highest, which would be a cheap source of flexibility for the DSO. However, the model struggles with limited flexibility for the shorter notice times, not being able to reallocate all of the displaced energy.

An implementation of Vehicle-to-grid (V2G) is investigated to see if the system can provide additional flexibility services in the IBDRP. The notice times of 2 and 12 hours are seen to not be sufficiently long, as V2G is seen to constrict the system even more. The 12 hour notice time is successful at delivering the requested power to the grid, with a 3.2% total cost increase at maximum power delivery, compared to BC. These results imply that it is essential to have a sufficiently long notice time if the DSO is to request any power to be exported to the grid through an IBDRP.

Lastly, the model is tried on an existing IBDRP which provides economic incentive through a percentage demand charge cost reduction (DCCR). The DCCR percentage varies based on the

notice time, and only the operational costs are considered. The IBDRP with a 12 hour notice time results in a 1.8% reduction in total operational costs, showing that the office building could profit from participating in the IBDRP, even in a scenario with an IBDR-signal activation every day.

8.1 Further Work

In this thesis, the real-time RH optimization framework used deterministic input data. It would be interesting to include a scenario picker into the model similar as to what was done in SDDP. A weighted forecast that predicts what hours is likely to have throttle activation could also be an interesting addition to the model. Implementing a different decomposition method than SDDP would also be interesting for comparison purposes.

When testing the model for different IBDRP-signals, only three notice times were considered, which were part of an already existing program. It would be interesting to perform a sensitivity analysis on which notice times the system can handle for different parameters. It would also be interesting to analyze the system for different activation hours. Also, the system investigated in this thesis was modeled as an aggregated load. It could be beneficial to try to model the system as a non aggregated EVPL. Although this would require a lot of data on each individual EV, the benefits could be a more realistic and detailed modeling.

Finally, it would have been interesting to investigate how the marginal cost curves for flexibility can be used as information for the DSOs in an IBDRP, to reflect the actual costs of the flexibility measures.

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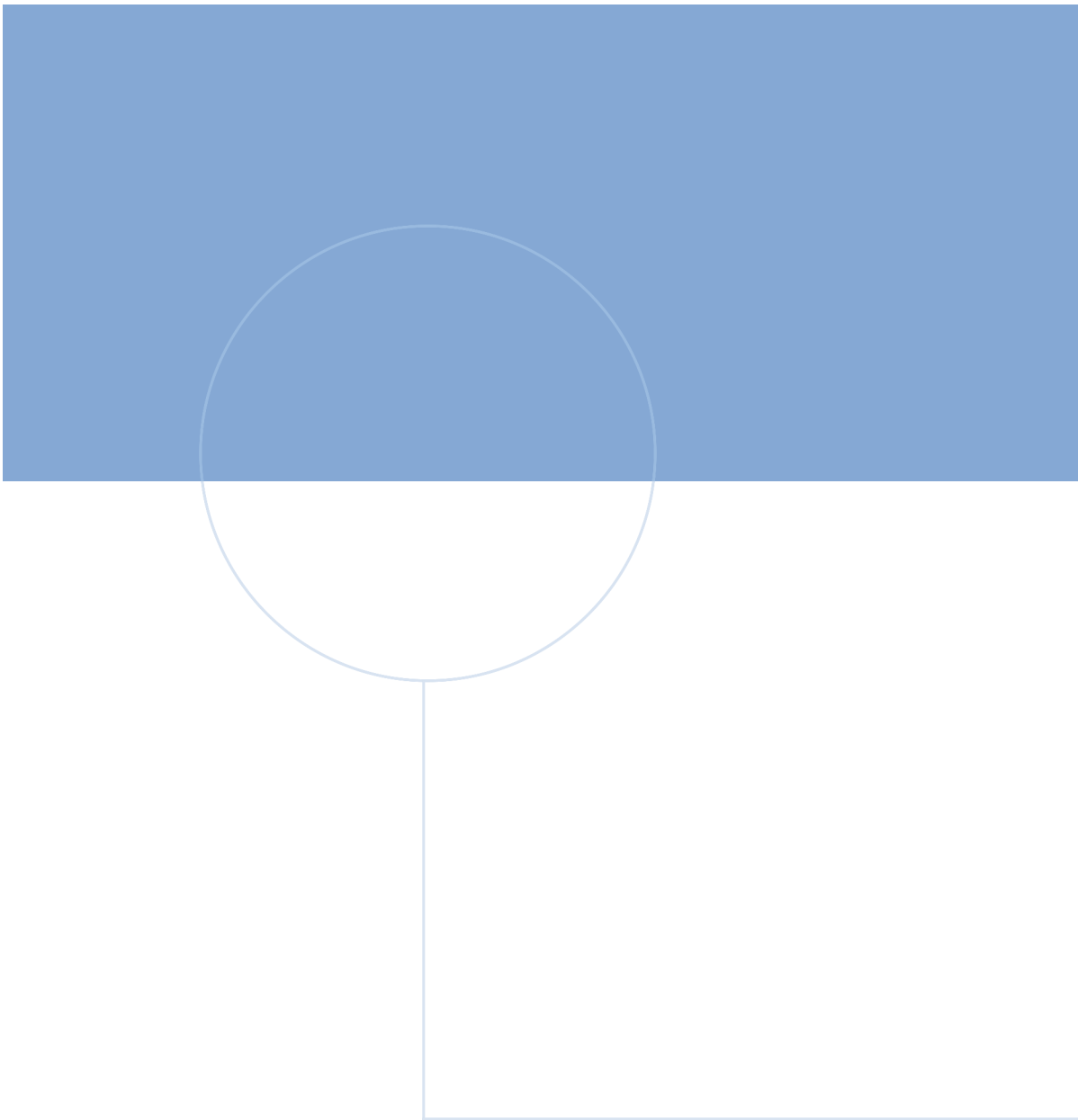
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